

Q&A

Please submit
questions at:
[https://aka.ms/
dataagent-qa](https://aka.ms/dataagent-qa)



A Practical Guide to Building and Deploying Fabric Data Agents for Deeper Insights

- **Wifi:**
 - Network: Fabcon26
 - Password: fabcon-26!
- **You should have received a paper slip with account information**
- **Please download content at:** aka.ms/fabcon-atl-2026-dataagent-tutorial
- **Open “Lab 1/Lab 1 – Getting Started.pdf”**
- **Begin the Lab**
- **We will begin the session at 9:15 AM**



A Practical Guide to Building and Deploying Fabric Data Agents for Deeper Insights

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Scott Votaw

Shreyas Canchi Radhakrishna

Intro the Speakers



Mayur Jain



**Shreyas Canchi
Radhakrishna**



Scott Votaw

Intro the Proctors



Sruthi Yalamanchili



Nellie Gustaffson



Misha Desai

Housekeeping



Submit all questions at aka.ms/dataagent-qa, and we will answer them during Q&A sessions throughout the tutorial.

- You can find this link by following the link on your account slips, and scrolling down to the Readme section.



Please utilize proctors! They are very knowledgeable about Data Agent and would be a great resource for question and guidance.

README

Data Agent Tutorial Fabcon EU 2025

This repository contains all the content required for the Data Agent Tutorial at Fab Con EU 2025.

Q&A During Tutorial

Please submit any and all questions during the tutorial to: aka.ms/dataagent-tutorial-qa

Learning Objectives



Build a working Data Agent over a dataset.



Understand key building blocks of Data Agent and how to leverage the building blocks to curate a working agent.



Learn how to evaluate and refine your Data Agent so it delivers value for its intended use case.



Learn how to consume your Data Agent through M365.



Learn how to about Data Agent's best practices for context.



Agenda – AM

Time	Topic
9:00 AM – 9:15 AM	Lab 1: Account Set up/Log In
9:15 AM – 9:45 AM	Introduction and Introduce Data Agent
9:45 AM – 10:15 AM	Data Agent Configurations and Architecture
10:15 AM – 10:45 AM	Morning Break
10:45 AM – 11:15 AM	Lab 2: Improve Data Agent Quality
11:15 AM – 11:45 PM	Data Agent SDK & Evaluation
12:00 PM – 1:30 PM	Lunch



Agenda – PM

Time	Topic
12:00 – 1:30 PM	Lunch Break
1:30 PM – 2:30 PM	Lab 3: Use the Evaluation Notebook to improve Data Agent quality
2:30 PM – 3:00 PM	Data Agent: Best Practices
3:00 PM – 3:30 PM	Afternoon Break
3:30 PM – 4:00 PM	Consumer Experiences & M365 Demo
4:00 PM – 4:30 PM	Lab 4: Unstructured Data
4:30 PM – 5:00 PM	Audit Logging

Data Agent – Intro

Data Agent in Microsoft Fabric

Deliver custom generative AI experiences for **your data** with **Data Agent**



Allow your users to talk to data in Fabric



Create, curate and configure a **data expert** on your data domain

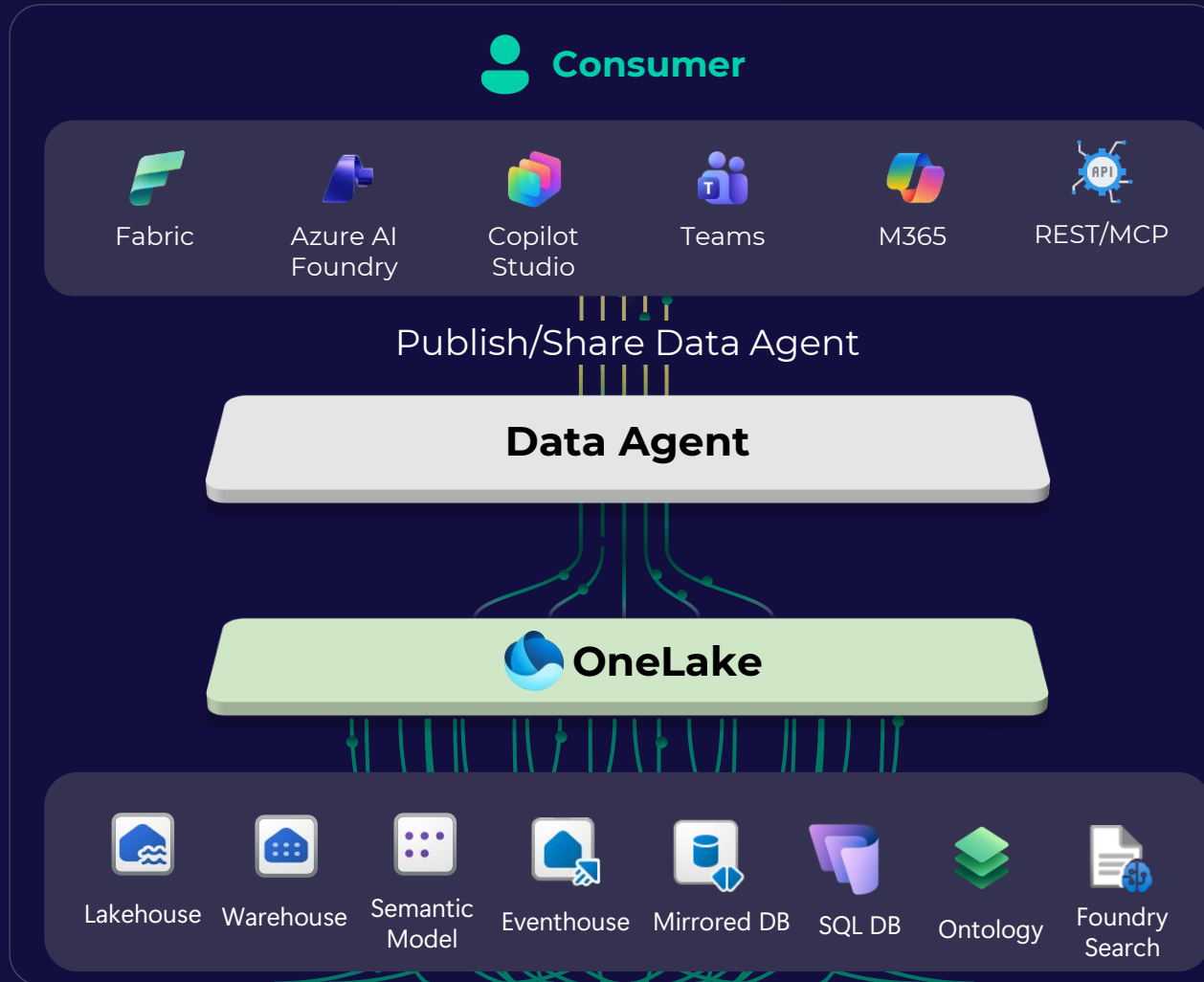


Share Data Agent with other creators and consumers in Fabric

The screenshot shows the Microsoft Fabric Personal-Workspace interface. The top navigation bar includes the Microsoft logo, 'Fabric', 'Personal-Workspace', and a search bar. The left sidebar contains icons for Home, Workspaces, OnLake, Monitor, Real-Time, Workloads, and Personal-Workspace. The main content area features a large circular icon with two overlapping squares and the text: 'Choose from predefined task flows or add a task to build one (preview)'. Below this, it says 'Select from one of Microsoft's predefined task flows or add a task to start building one yourself.' There are two buttons: 'Select a predefined task flow' and 'Add a task'. At the bottom, there is a table with the following data:

Name	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
AdventureSales-DataAgent-Scenario	Data agent	—	Amir Jafari	—	—	—	Confidential(Mi...	ⓘ

Data Agent in Microsoft Fabric



This **conversational data agent** allows users to interact with a virtual analyst



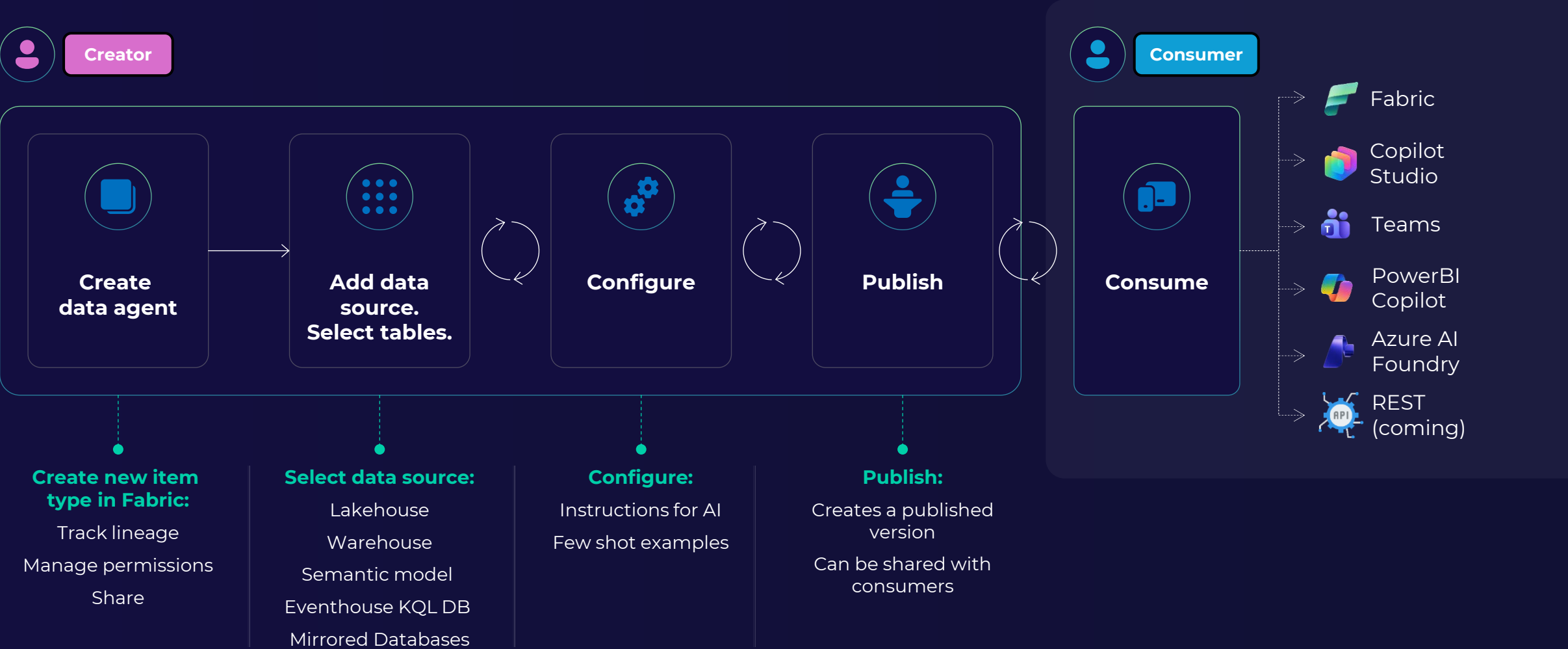
Seamlessly reason over various Data Sources such as SQL Databases in Fabric and other sources to create a **powerful data expert tailored to your data domain**



Data agents can be consumed inside and outside of Fabric and used as knowledge sources for **custom agents in Copilot Studio and Azure AI Foundry**

Data Agent end-to-end flow

End-to-end scenario



Data Agent Users



Creator

- We will take on this role during this tutorial
- Responsible for Configuring the Data Agent

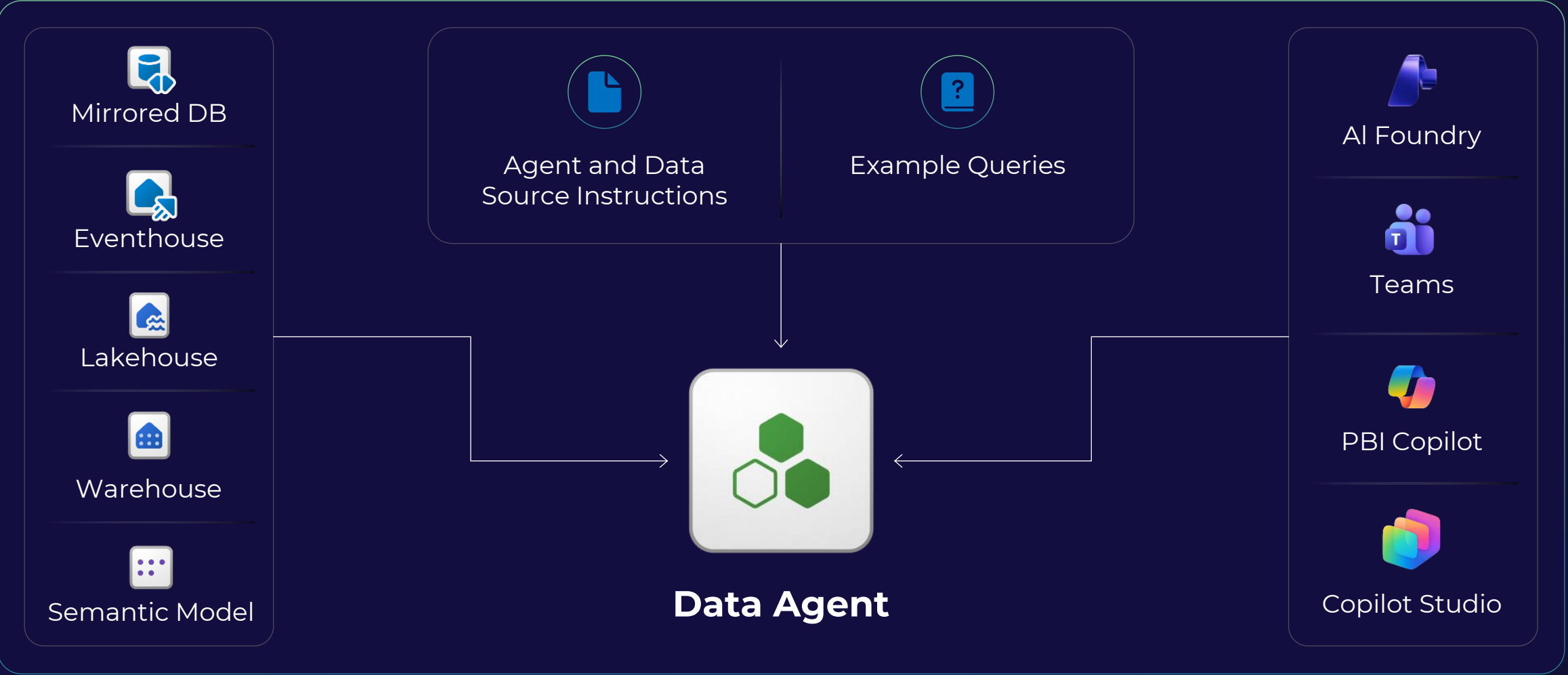


Consumer

- Consumers of the Data Agent
- They are the users that will ask your data agent questions
- Note: They may not have knowledge on how the Agent is configured.

Data Agent – Getting Started

Creating a Data Agent

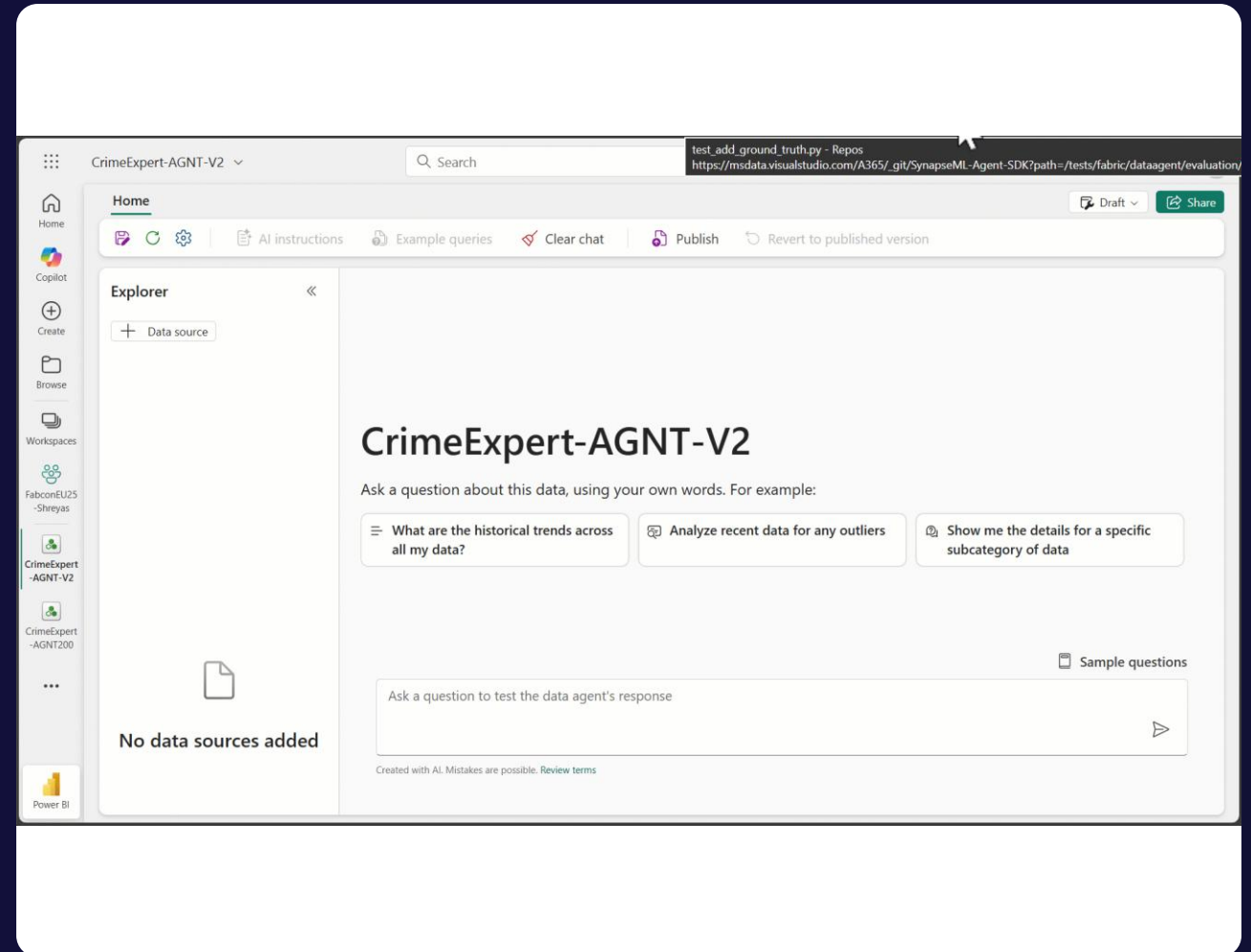


Schema Selection

Select Datasources out of a wide variety of artifact types:

- Lakehouses
- Warehouses
- Mirrored Artifacts
- Eventhouse KQL
- Semantic Models

Select Schema Elements



Schema Access with User Credentials



Data Agent adheres to Fabric One Lake's security



The system first uses the credentials of the Creator to access the schema of the data source

- This ensures that the system fetches data structure information that the user has permission to view
-



Consumers are required to have access to tables that Data Agent will use as an input

Data Agent Architecture & Configurations

AI Architecture

- Choose tools
- Only general knowledge
- Rephrase
- Orchestrate
- Summarize/Format

Orchestrator

**Conversation
History**

Specific knowledge/capabilities

Tools

Data Agent Benefits

Orchestrator

- Compliance
- Security
- Maintenance
- Extensibility
- Support

Conversation History

- Compliance
- Security

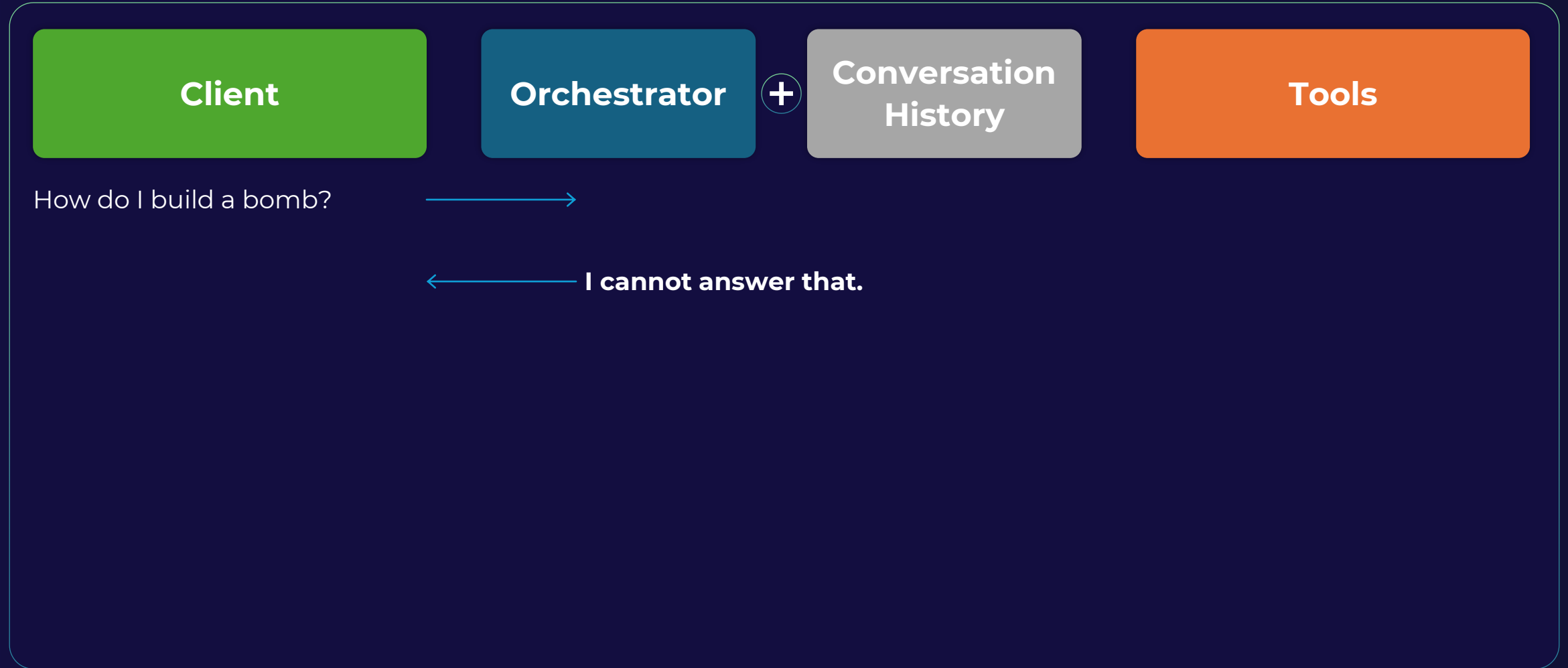
Tools

- Auto-generation
- Fabric-native performance
- Maintenance
- Security
- Extensibility

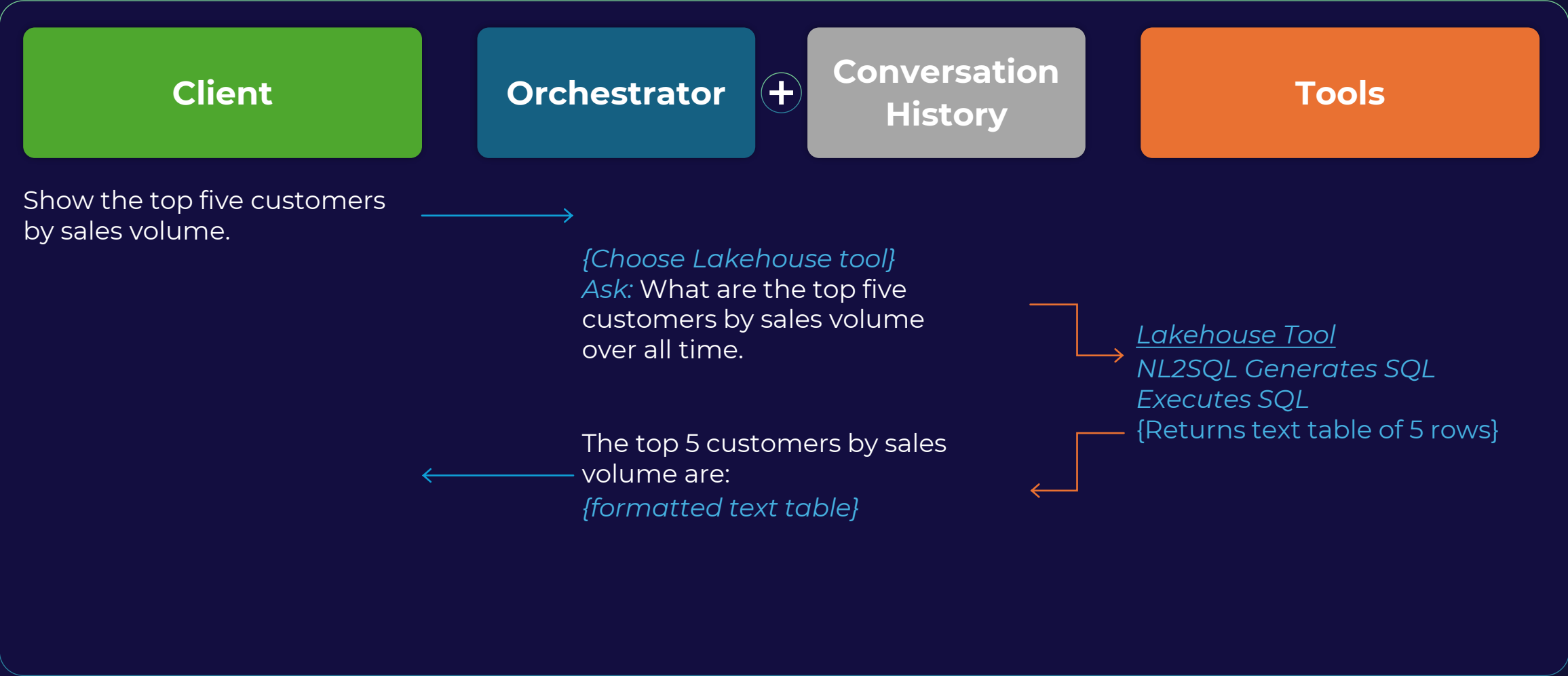
Question flow (no need for tools)



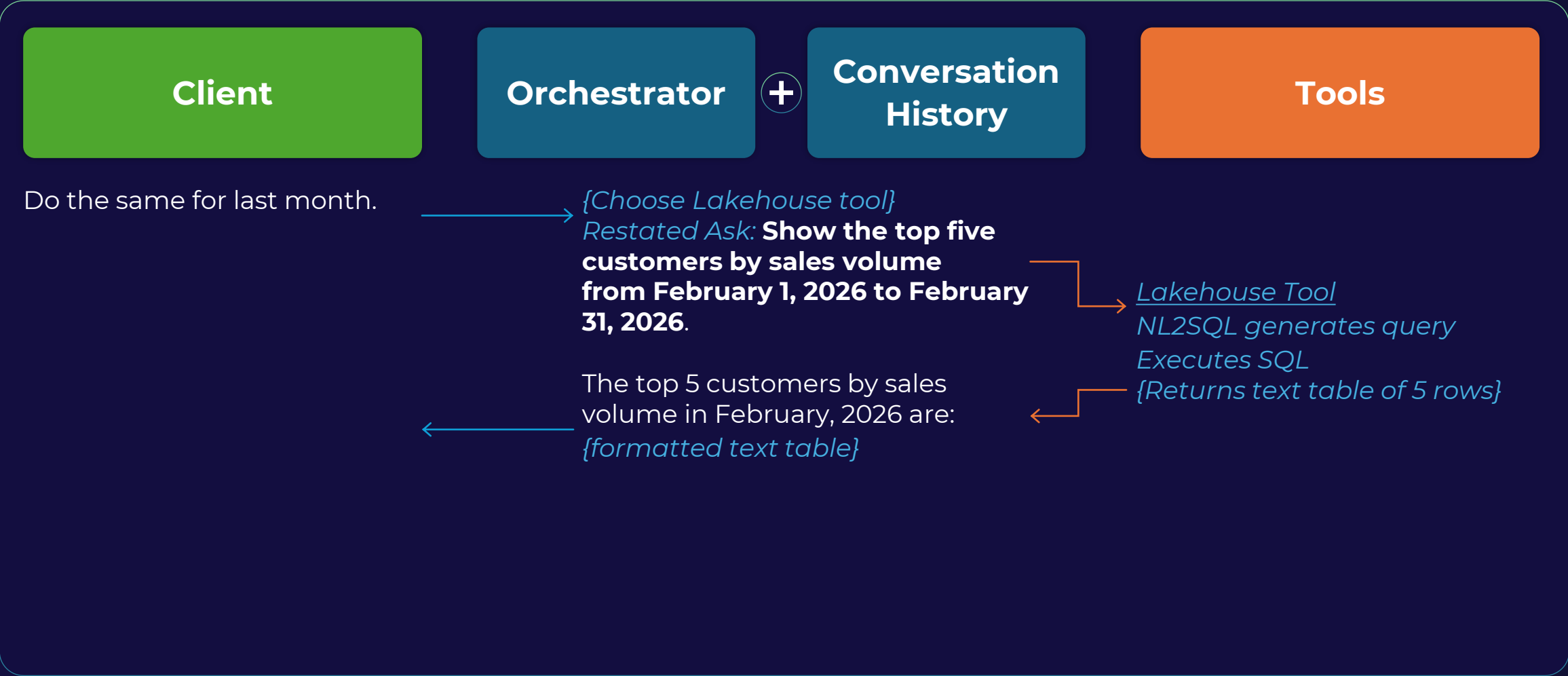
Question flow (RAI)



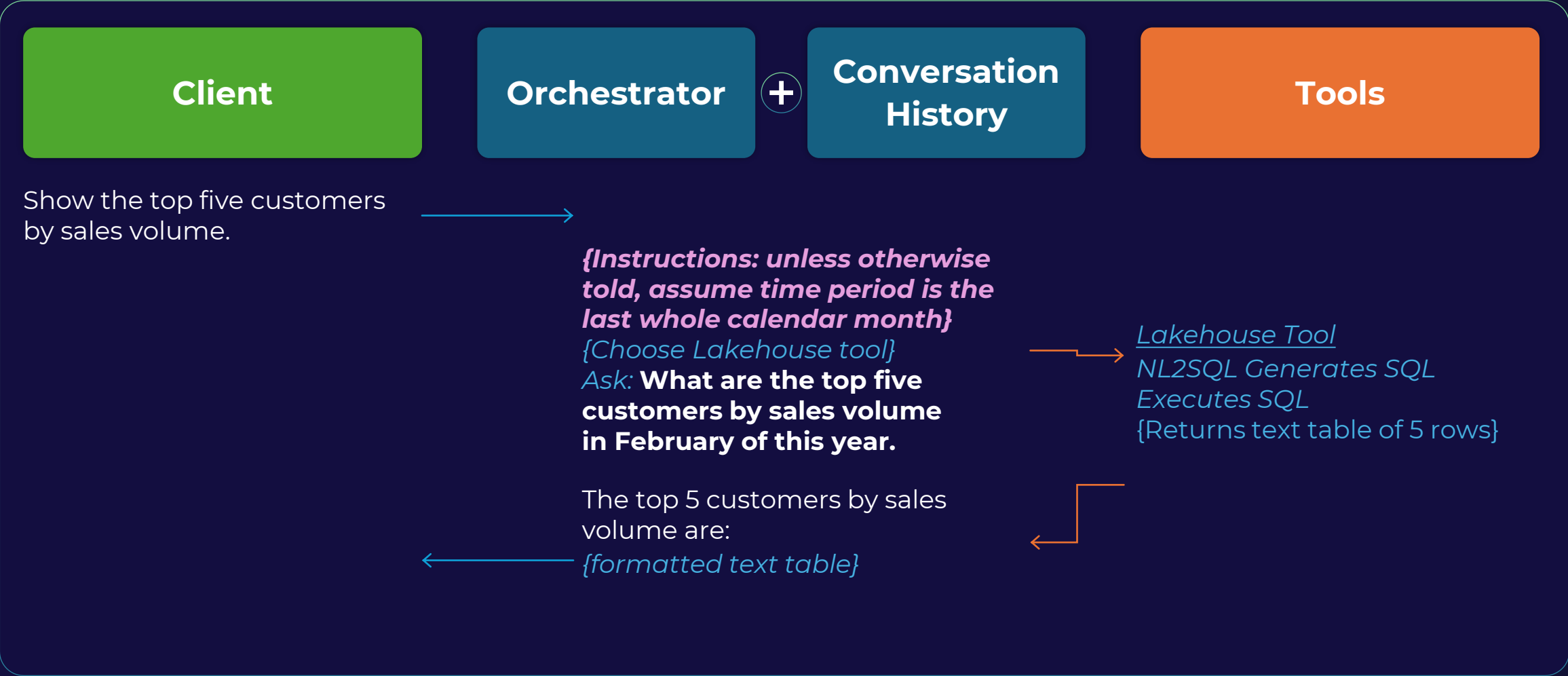
Question flow (canonical single tool flow)



Question flow (question with history)



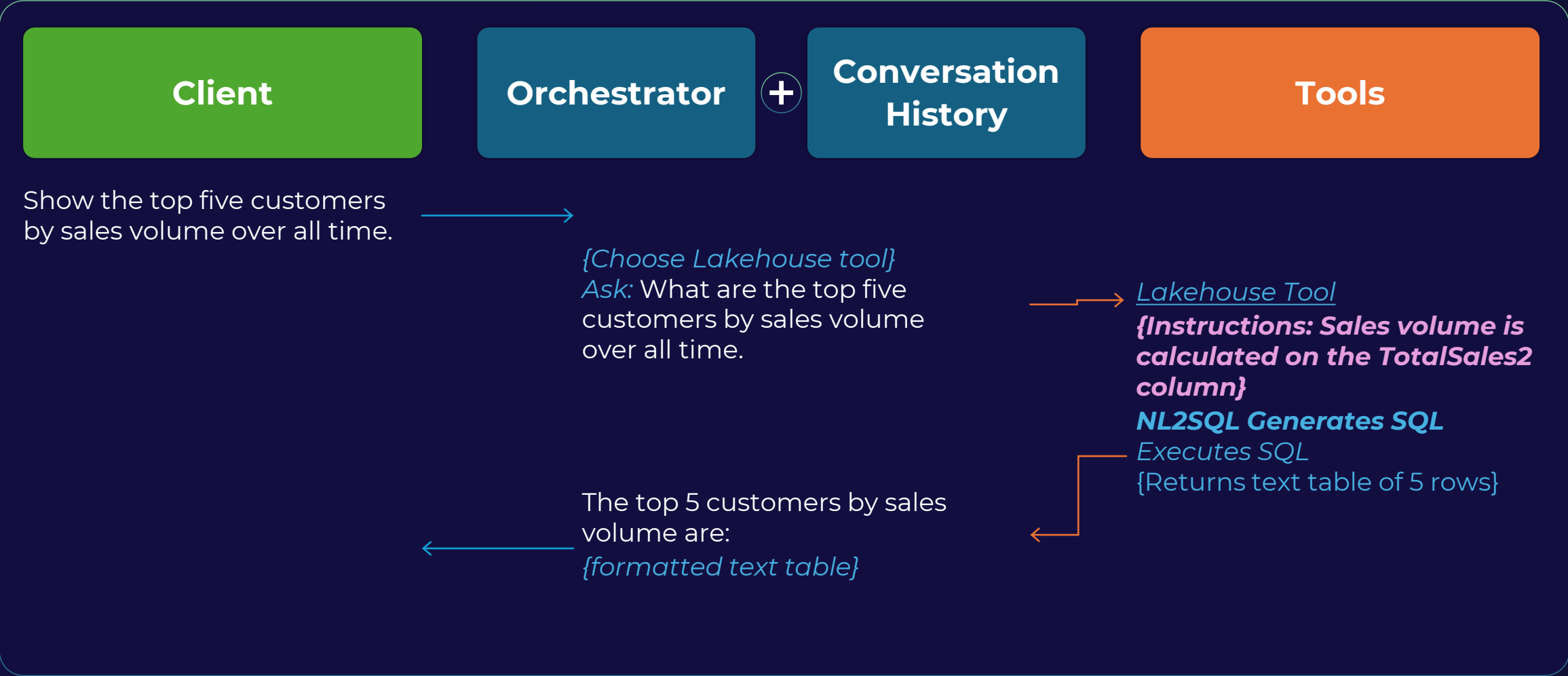
Question flow (agent/orchestrator instructions)



Question flow (agent/orchestrator instructions)



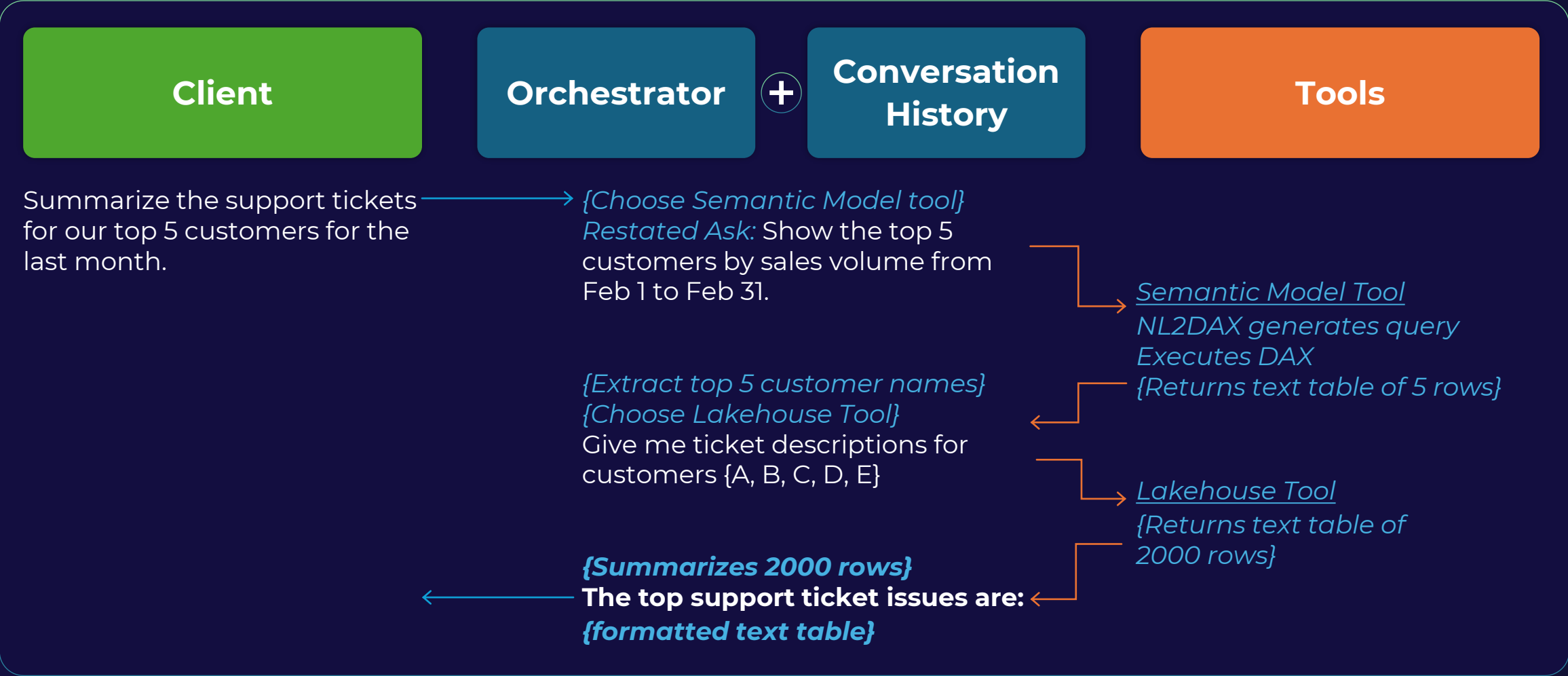
Question flow (data source instructions)



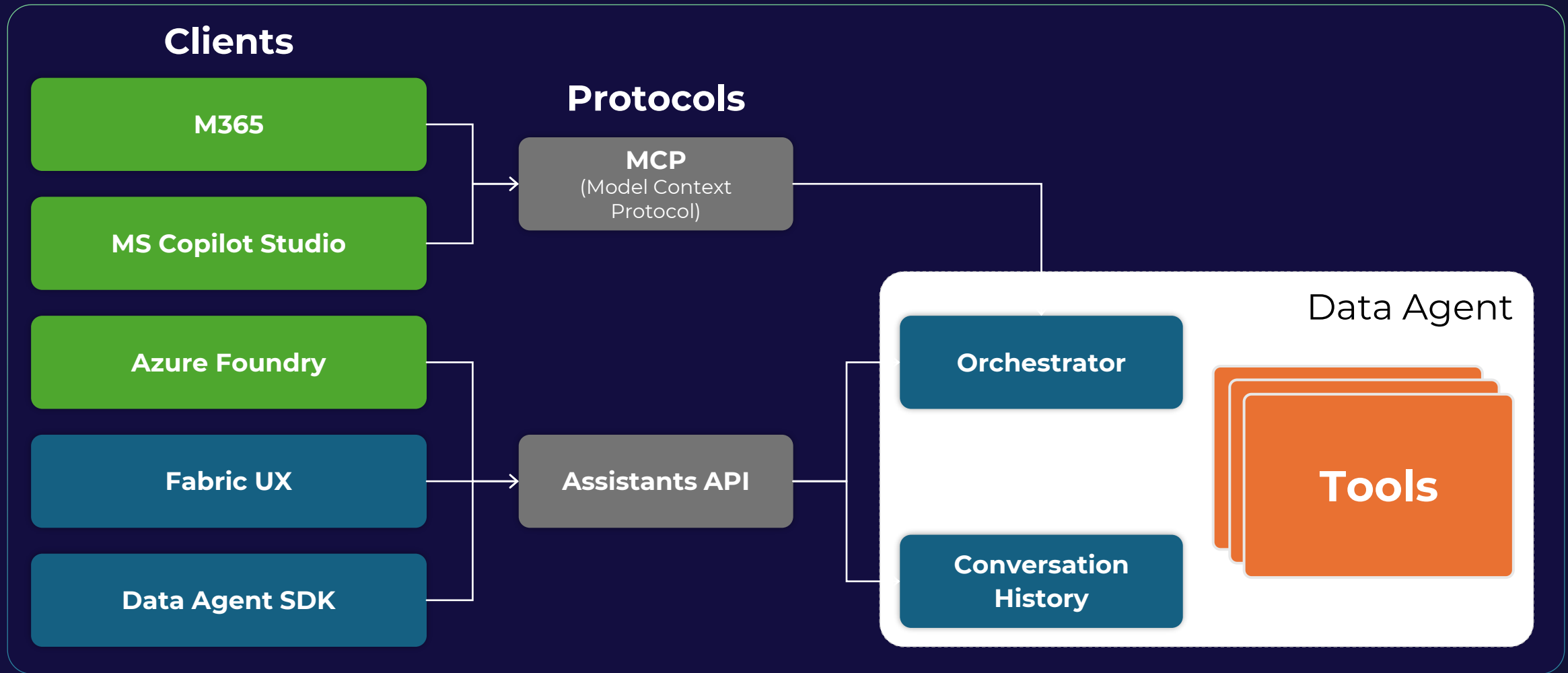
Question flow (question with answer in history)



Question flow (multi-tool)



Data Agent Architecture – Protocols



Double Orchestration

All have **Orchestrator**
and **Conversation
History**

M365

MS Copilot Studio

Azure Foundry

Orchestrator

Conversation
History

Data Agent

Tools

Example SQL queries

Add examples of how a natural language question would become a SQL query for your data. The data agent will automatically select up to three valid examples to share with the model each time it handles a user question. [Learn more](#)

+ Add example ↑ Import from .json ↓ Download all as .json 🗑 Delete

🔍 Filter by keyword

Showing 1 items (0 selected)

☐ Question

SQL query

☐

Which Streets in Manchester have the highest crime rates?

59/500 characters used

```
SELECT Location AS StreetName, COUNT(*) AS NumberOfCrimes
FROM [GreaterManchesterCrime].[greatermanchestercrime]
-- note that we ALWAYS want to perform case insensitive searches
```

368/1000 characters used

Assistant APIs



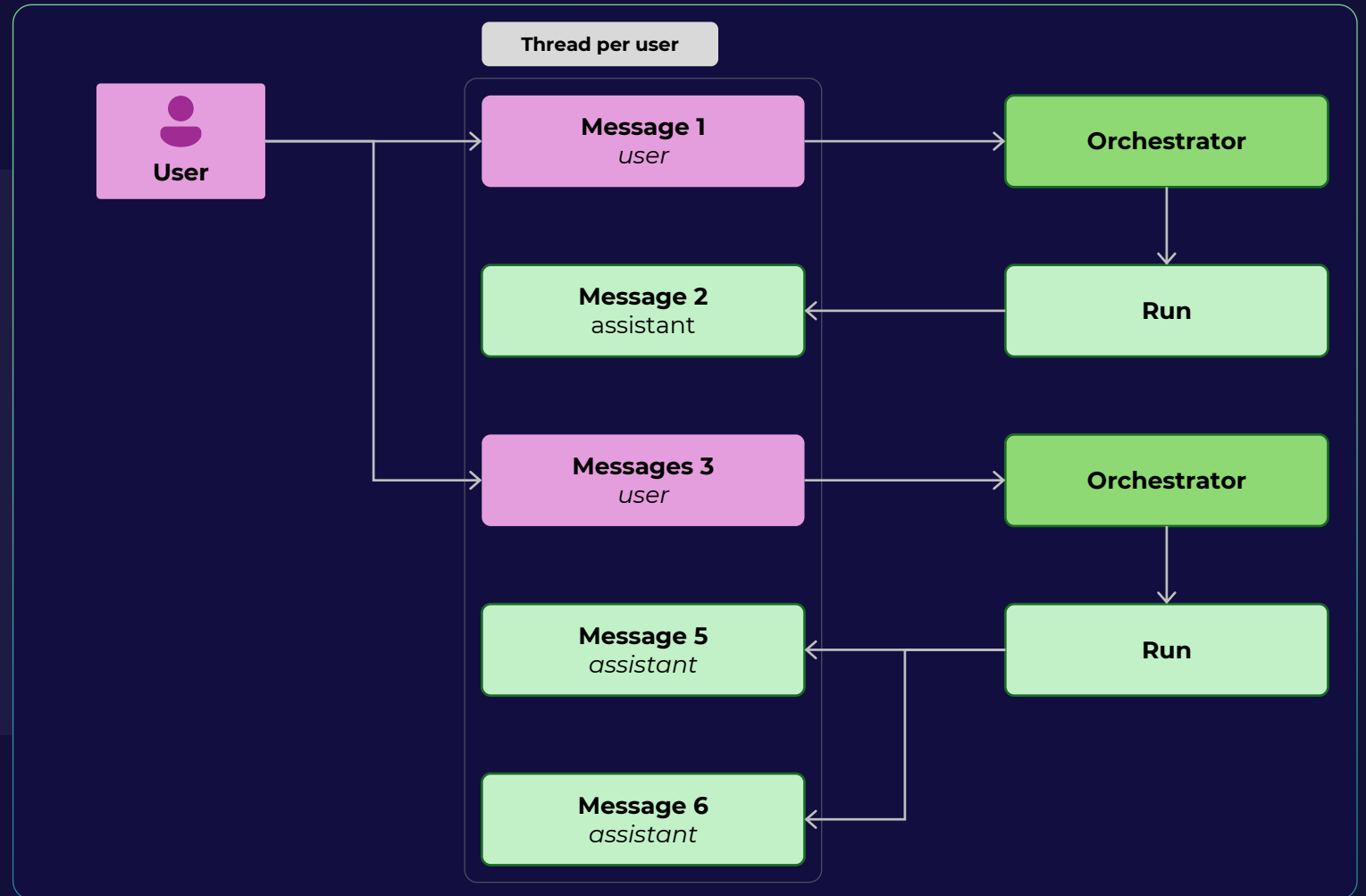
Messages



Threads



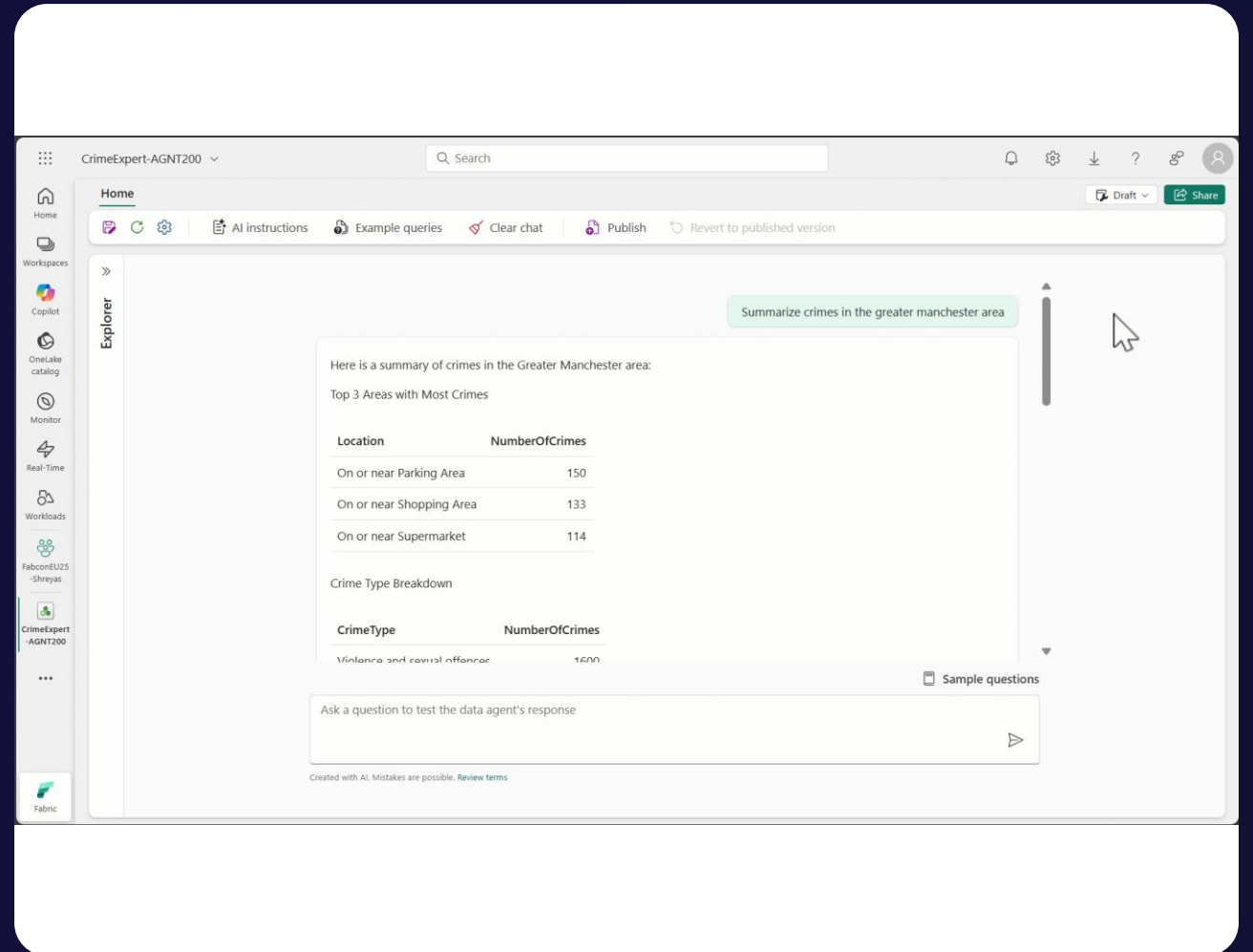
Runs



Data Agent UX: Run Steps

Data Agent displays run steps:

- Tool that was chosen
- The query executed
- The query output



Thread

Message 1:

Submitted by the user

Where is the place with the largest number of violent crime events?

The place with the largest number of violent crime events is "On or near Parking Area," with 38 such events.

1 step completed ^

Response time: 12 sec

✓ Analyzed ShreyasFabconEU25Data Lakehouse for: "Find the location with the highest number of violent crime events and provide the number of such events."

</> Query code

Sql

```
SELECT TOP 1 Location, COUNT(*) AS NumberOfViolentCrimes
FROM [GreaterManchesterCrime].[greatermanchestercrime]
WHERE [Type] = 'Violence and sexual offences'
GROUP BY Location
ORDER BY NumberOfViolentCrimes DESC
```

Query output

Location	NumberOfViolentCrimes
On or near Parking Area	38

Sample questions

Ask a question to test the data agent's response

User Input

Created with AI. Mistakes are possible. [Review terms](#)

Message 2:

Assistant Generated Response based on Run steps

Run steps

generated by Data Agent

Home

AI instructionsExample queriesClear chatPublishRevert to published version

Explorer

Restated Question

3 steps completedResponse time: 9 sec

Analyzed Kaggle Lakehouse for: "List the top 3 areas (from the Location column) where crimes occur most frequently in the Greater Manchester area including the count."

Query code

Sql

SELECT Location, COUNT(*) AS NumberOfCrimes
FROM [GreaterManchesterCrime].[greatermanchestercrime]
GROUP BY Location
ORDER BY NumberOfCrimes DESC
OFFSET 0 ROWS FETCH NEXT 3 ROWS ONLY

Query output

Location	NumberOfCrimes
On or near Parking Area	150
On or near Shopping Area	133
On or near Supermarket	114

Analyzed Kaggle Lakehouse for: "Provide a breakdown of the count of each crime type"

Sample questions

Ask a question to test the data agent's response

Created with AI. Mistakes are possible. Review terms

Orchestrator Context

AI instructions

Agent instructions

Guide the data agent to the best data sources for different types of questions by writing a detailed system prompt.

When displaying results, I want you to always use table output for tabular results in Markdown format.

When asked about areas, the user intends to use the location column in the GreaterManchesterCrime dataset.

When used to summarize crimes in the greater manchester area:
1. Find Top 3 areas where crimes occur.
2. Break down the count based on the crime type
3. Identify the Isoa that has the most crimes

461/15000 characters used

Data source instructions

For each connected data source, help the data agent understand its data and how to use it most effectively.

Kaggle instructions

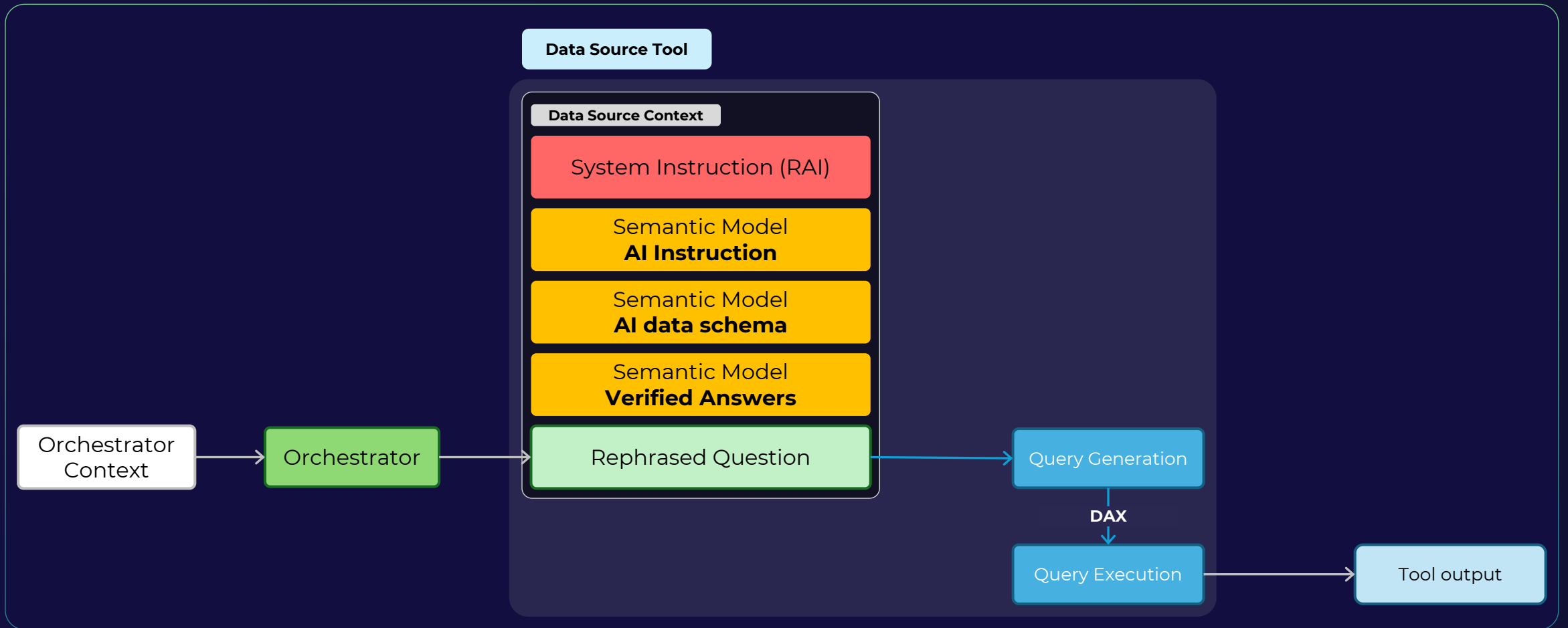
Table descriptions
Below are the possible values in the type column:
- Violence and sexual offences
- Public order
- Criminal damage and arson
- Vehicle crime
- Burglary
- Other theft
- Shoplifting

298/15000 characters used

Selected Schema

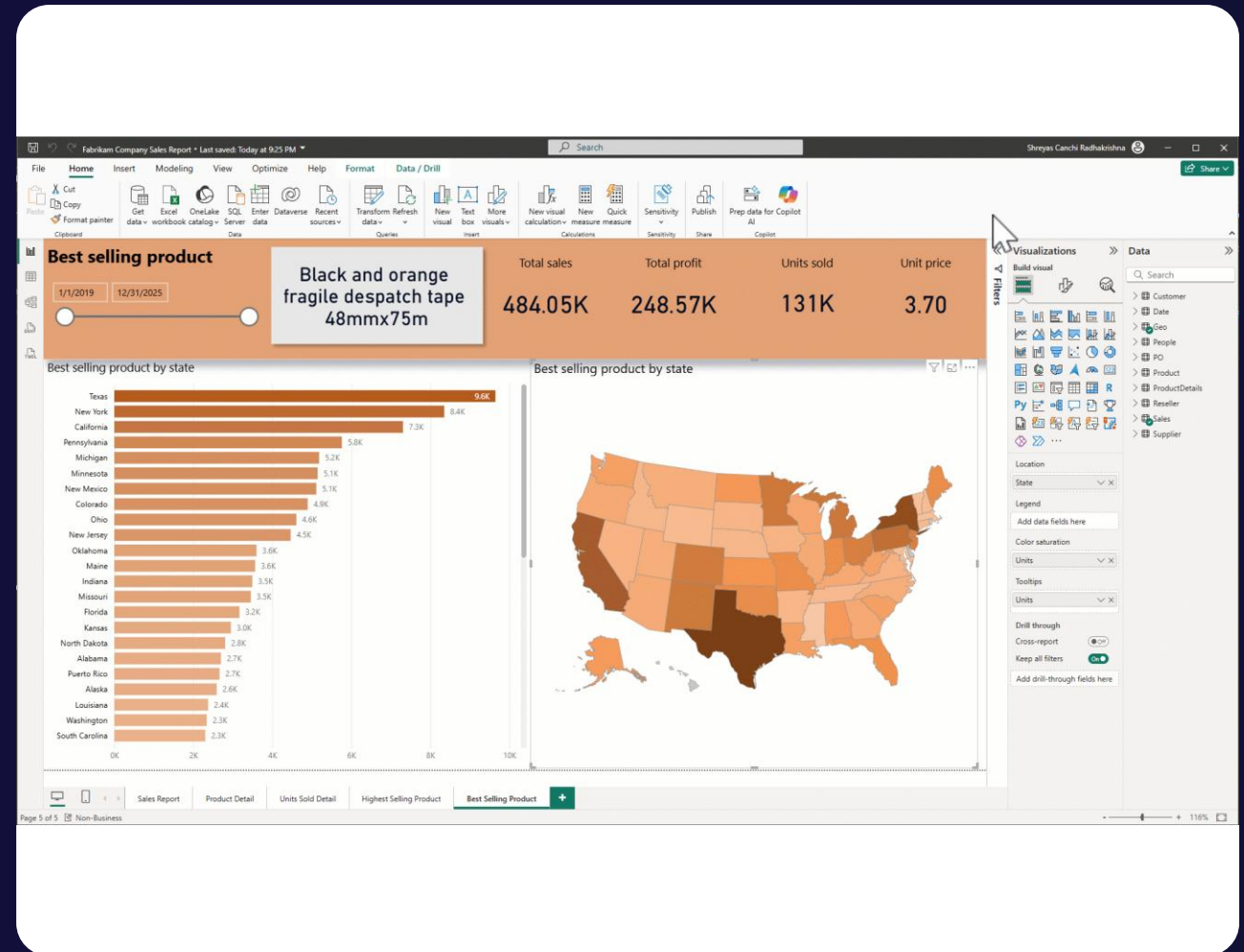
Data Source Tool Output

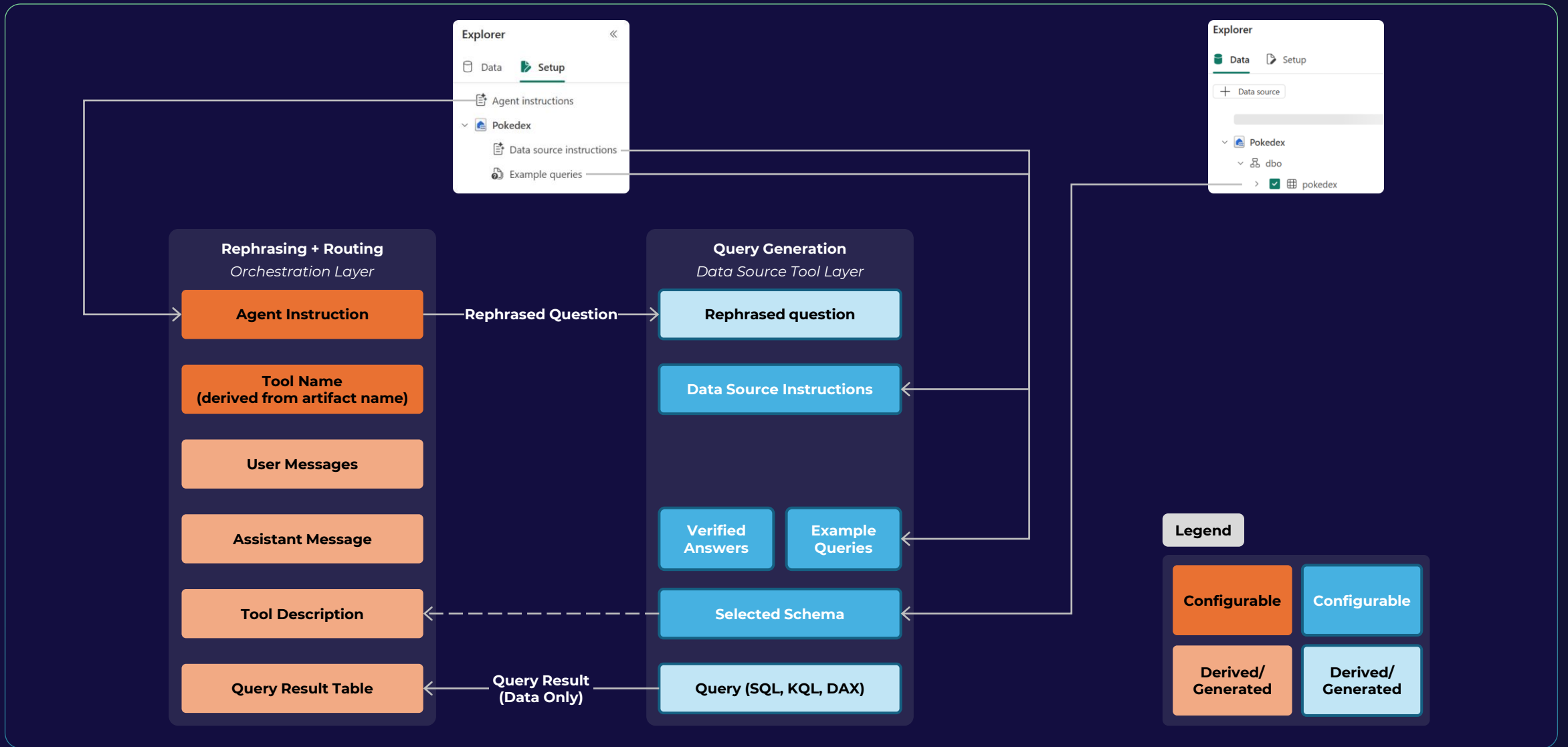
Data Source Tool: Semantic Model



Configure Semantic Models for AI

- Configure your Semantic Model in PBI Desktop
- Update Instructions, Verified Answers, and Schema Selection





Creator Configuration

Agent Instructions

Intended Use Cases:

- Specify which data sources to route for specific questions
- Guide how the agent interprets user intent

What to include:

- Map terminology to user intent
- Define steps for handling key questions
- Set response style and tone

Data Source Instructions

Intended Use Cases:

- Provide clarity on how to use each source
- Guide table selection for topics
- Ensure correct query logic

What to include:

- Descriptions of key tables & columns
- Recommended table usage for certain topics
- Example column values and filter patterns

Example Queries

Intended Use Cases:

- Data source instruction overflow (RAG)
- Reflect advanced logic
- Examples should highlight join patterns or expected values

What to include:

- Representative set of questions a consumer may ask

Q&A

Please submit
questions at:
[https://aka.ms/
dataagent-qa](https://aka.ms/dataagent-qa)



Lab 2: Learn to Configure a Data Agent

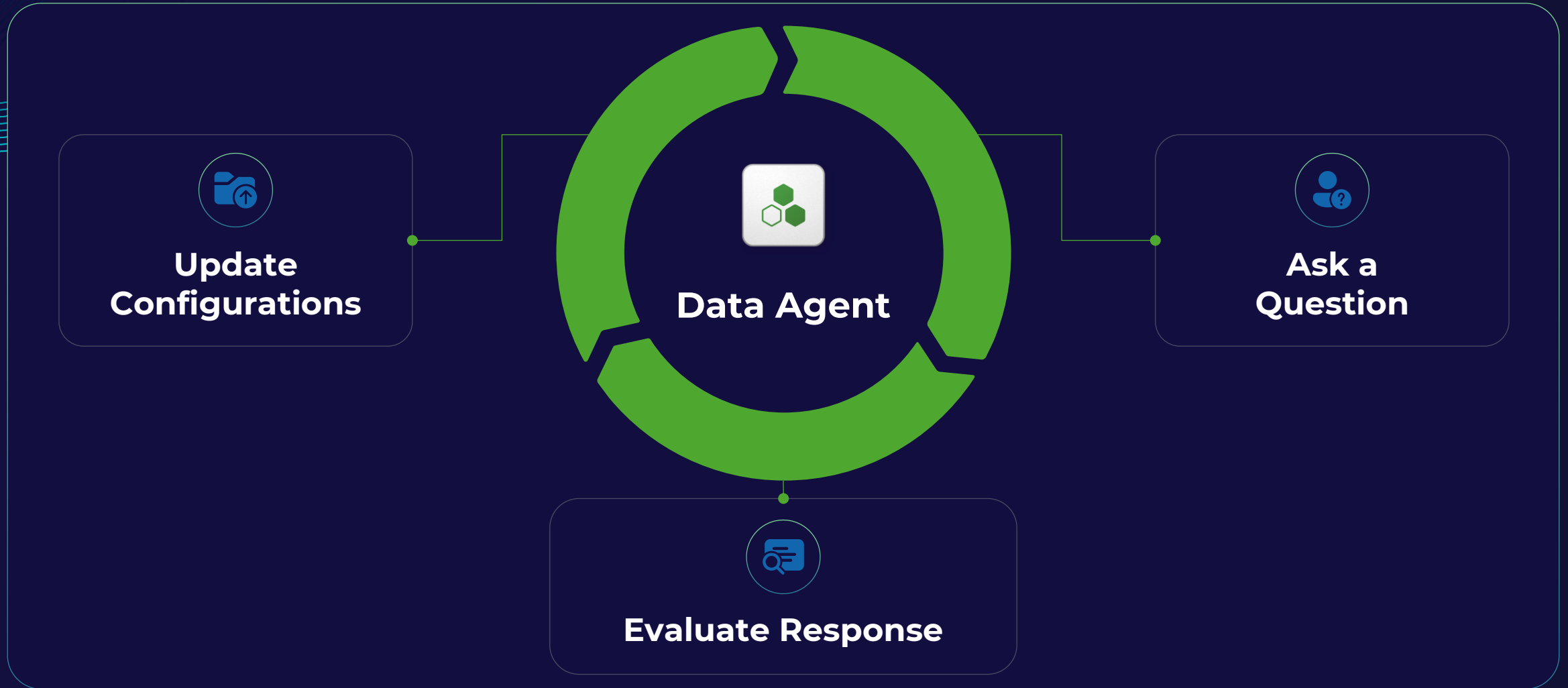
Q&A

Please submit
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dataagent-qa](https://aka.ms/dataagent-qa)



Data Agent SDK and Evaluation

Building a Data Agent is Iterative

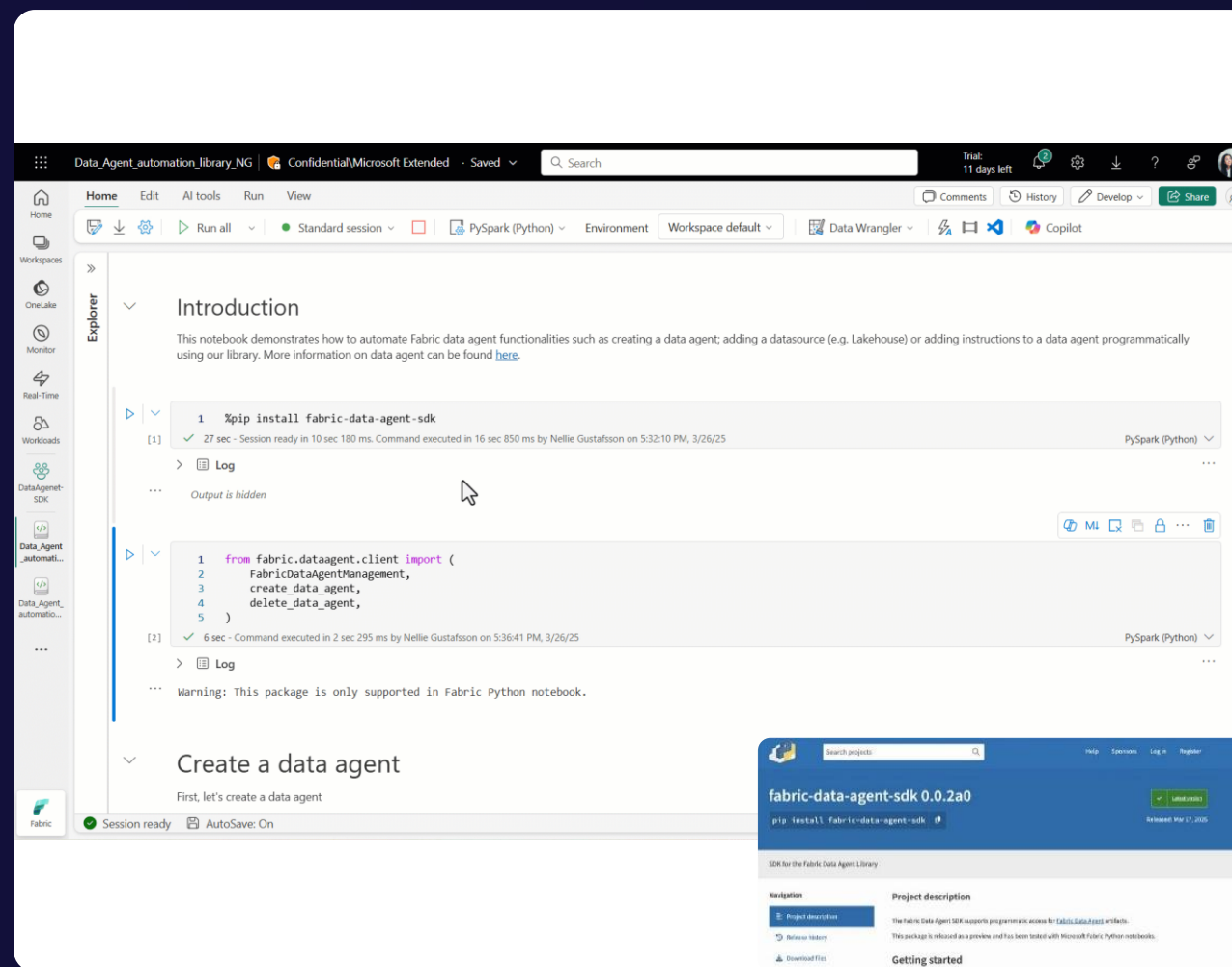


Fabric data agent SDK

Leverage the **Data Agent SDK** to invoke Fabric data agents from Fabric Notebooks

Create and configure data agents for automation and testing

Test and evaluate data agents against your ground truth datasets



Evaluate data agents

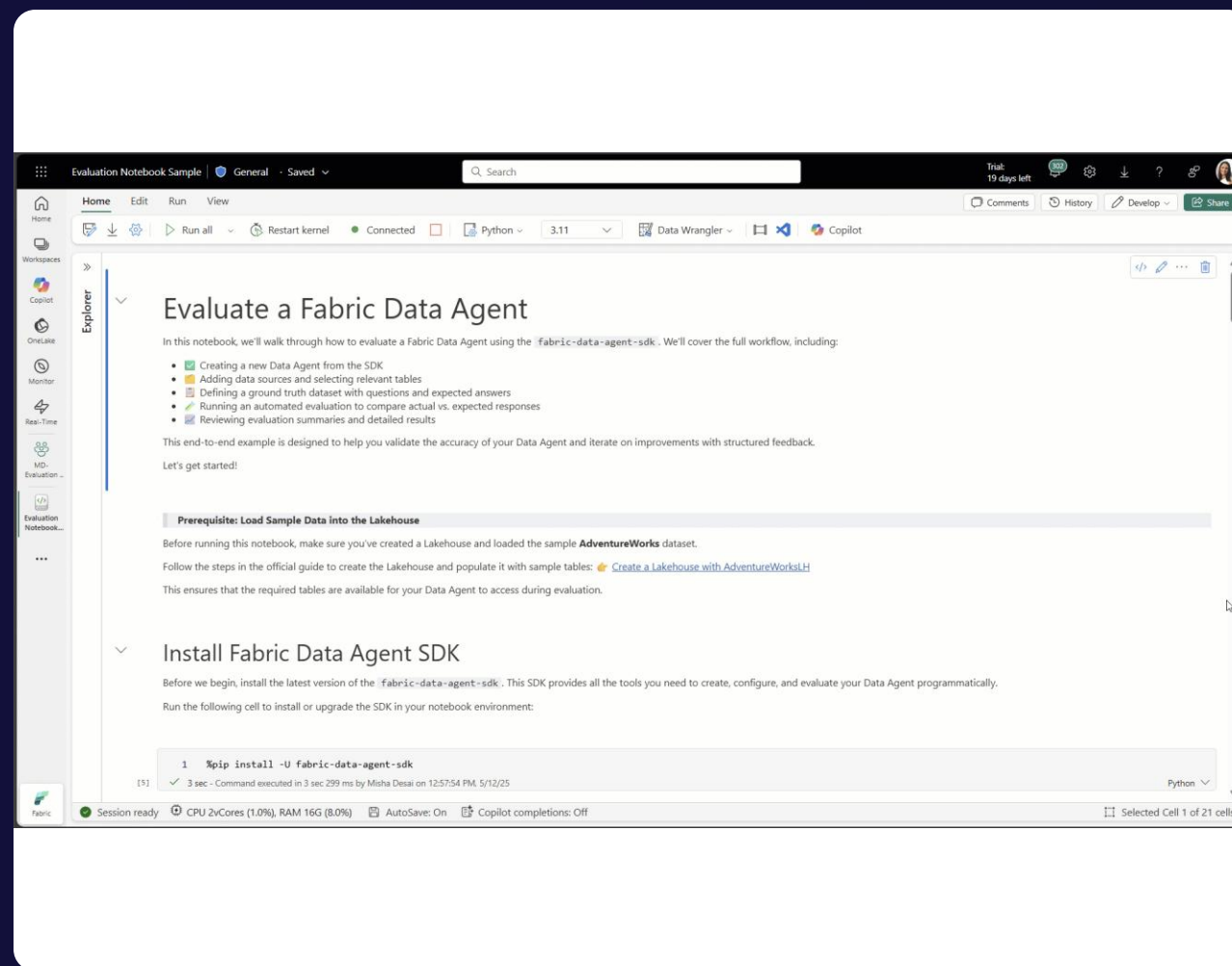
Native support for **data agent evaluation** through the Fabric data agent SDK

Run structured evaluations using Python from notebooks

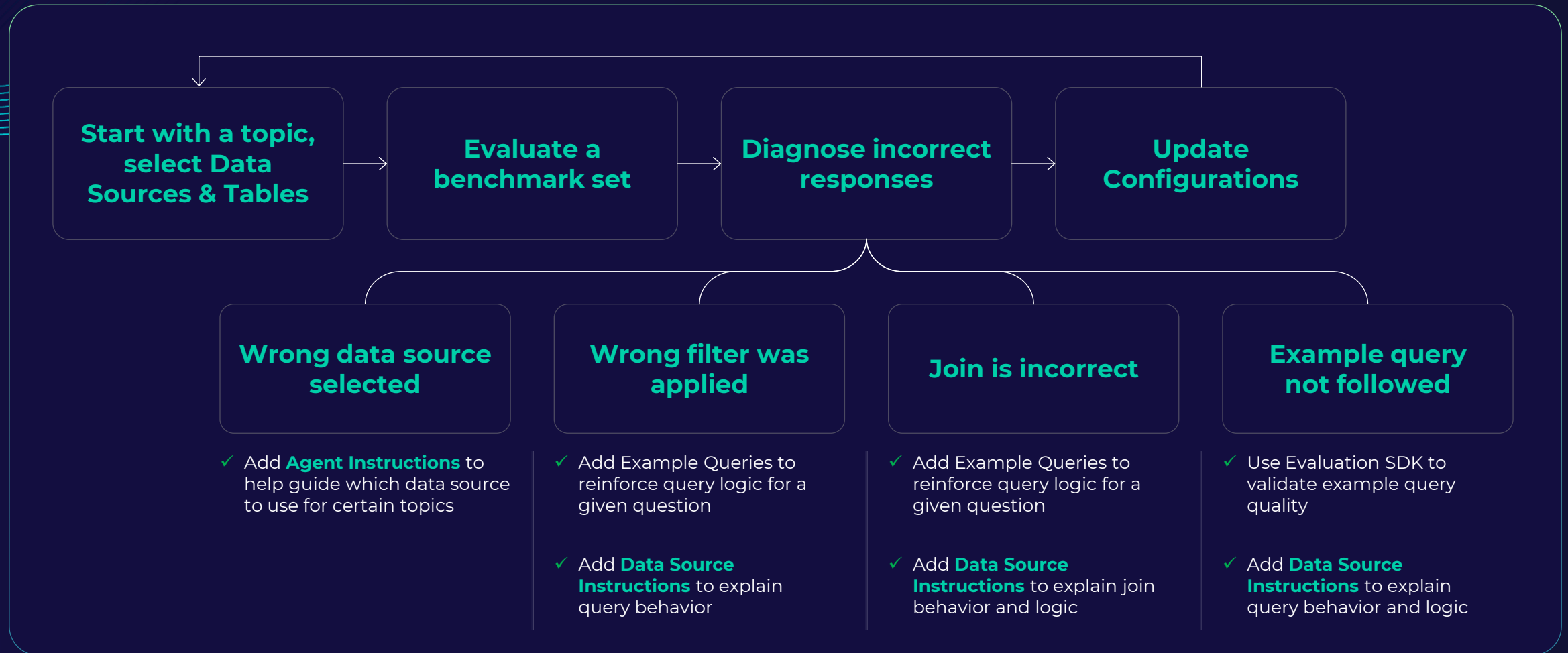
Validate accuracy before production deployment

Tune prompts for better performance

Benchmark improvements over time



Configuration Flow – Evaluation SDK



Generating a Benchmark



Understand the Consumer – what questions will they likely ask?



Clear and concise question + ground_truth pairs



Use aggregates rather than large outputs

- “List all product categories and their revenue in 2017” → “What are the top 3 Product Categories by Revenue”



Questions



Ground_truth

Give me the revenue from furniture related purchases

The revenue for furniture related purchases is **\$1,143,364.98**

What was the revenue from health and beauty related purchases in 2017?

The Revenue from health and beauty related products purchased in 2017 is **\$481,755.71**

Compute the median product price for category beauty and health.

The median price for health and beauty products is **79.9**

Evaluation Framework

Benchmark

Question	Ground_truth
What type of...	The type of cri...
...	..
...	...
...	...



Evaluation
Framework



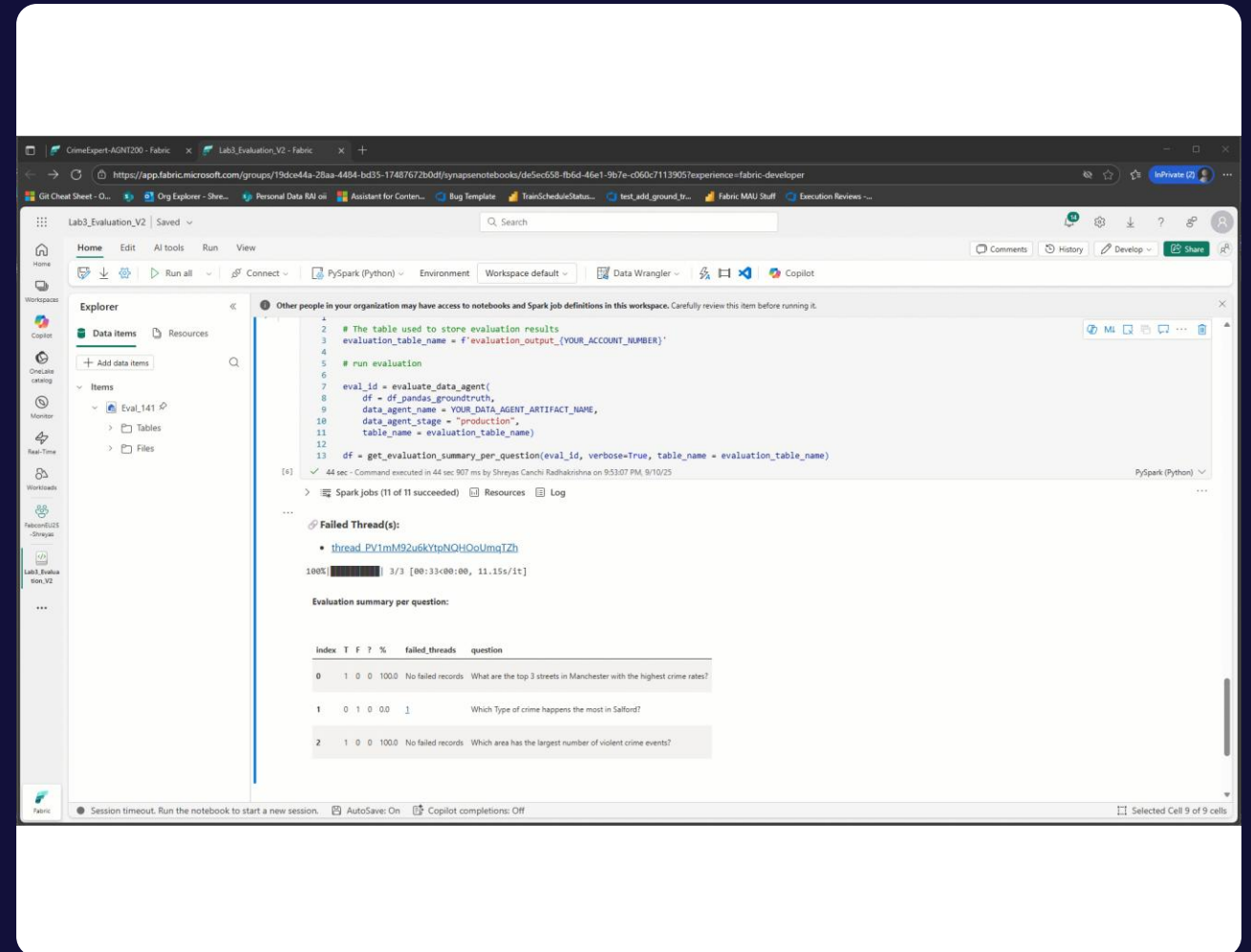
Evaluation Result

Result	Failed Thread	Question
Pass	N/A	What type of...
Fail	url	...
...	...	...
...	...	...

Diagnosing Incorrect Responses

Evaluation failures can be caused by multiple reasons

- Configs could improve
- Incorrect groundtruth
- Evaluator made a mistake
- Confusing Column names



🔗 The result does not match the provided ground truth(s):

- [thread_77StYhqjKnIkPtKr7uB2G1SU](#)
- [thread_JIBrWUvJIt3fg3p7xdZtEp7V](#)
- [thread_xBaitmKoJyXFnuasoDWuV10R](#)
- [thread_1AMW6dPbszUq6vy8uOcj5Bkj](#)

100% | ██████████ | 7/7 [00:52<00:00, 7.49s/it]

Evaluation summary per question:

index	T	F	?	%	failed_threads	question
0	0	1	0	0.0	1	Compute the median product price for category beauty and health.
1	1	0	0	100.0	No failed records	Find the top city by the number of delivered orders.
2	0	1	0	0.0	1	Give me the revenue from furniture related purchases
3	0	1	0	0.0	1	Please provide the average, median, p95, and p99 for the delivery time of the City with the most number of deliveries.

If a Thread has failed, you can find the link to the failed thread here.

Result
Summary

index	question	expected_answer	actual_answer	evaluation_judgement	thread_url
0	Compute the median product price for category beauty and health.	The median price for health and beauty products is 79.9	No data was found for the median product price in the "beauty and health" (beleza_saude) category. This may mean there are no sales records with price data for this category, or that the price column is not available or named differently in the database. If you have another category or would like to adjust the search, please let me know!	False	thread_77StYhqiKnkPtKr7uB2G1SU
1	Give me the revenue from furniture related purchases	The Revenue for furniture related purchases is 1143364.98000001	The closest matching product category to "furniture" is furniture_decor. The total revenue from furniture related purchases is \$902,511.79. Product Category Total Revenue ----- ----- -----: furniture_decor \$902,511.79	False	thread_JIBrWUvJlt3fg3p7xdZtEp7V
2	Please provide the average, median, p95, and p99 for the delivery time of the City with the most number of deliveries.	Zip Code: SAO PAULO Average: 7.97 Median: 7 P95: 19 P99: 31	Here are the delivery time statistics for the city with the most deliveries (SAO PAULO): Statistic Value (Delivery Days) ----- ----- -----: Average 7 Median 7 95th Percentile 19 99th Percentile 31 These values are based on the Delivery Days for orders delivered in SAO PAULO.	False	thread_xBaitmKoJyXFnuasoDWuV10R
3	What was the revenue from health and beauty related purchases in 2017?	The Revenue from health and beauty related products purchased in 2017 is 550420.110000003	The total revenue from health and beauty related purchases in 2017 was 481,755.71.	False	thread_1AMW6dPbszUq6vy8uOcj5Bkj

Diagnosing the Issue

No data was found for the median product price in the "beauty and health" (beleza_saude) category. This may mean there are no sales records with price data for this category, or that the price column is not available or named differently in the database. If you have another category or would like to adjust the search, please let me know!



2 steps completed ^

Response time: 13 sec

⊗ Analyzed ecommerce_lakehouse Lakehouse for: "Compute the median product price for the product category 'beleza_saude' using the relevant product price or sales columns."

</> Query code

Sql

```
SELECT TOP 1
  P.[Product Category English] AS ProductCategory,
  PERCENTILE_CONT(0.5)
    WITHIN GROUP (ORDER BY S.Price)
    OVER(PARTITION BY P.[Product Category English]) AS MedianPrice
FROM
  [fda].[Sales] S
INNER JOIN
  [fda].[Products] P ON S.[Product ID] = P.[Product ID]
WHERE
  P.[Product Category English] = 'beleza_saude' AND
  S.Price IS NOT NULL
```

⌘ Query output

Failed to execute step (RAID: 37a1be78-a37a-44c7-8129-0be1c1804e4a). Error: No data found after query execution.

lakehouse for: "Find the exact product category key(s) in fda.Products that match or 1."

lakehouse for: "Compute the median product price for the product category product price or sales columns."

**Incorrect
Filter due to
rephrased question**

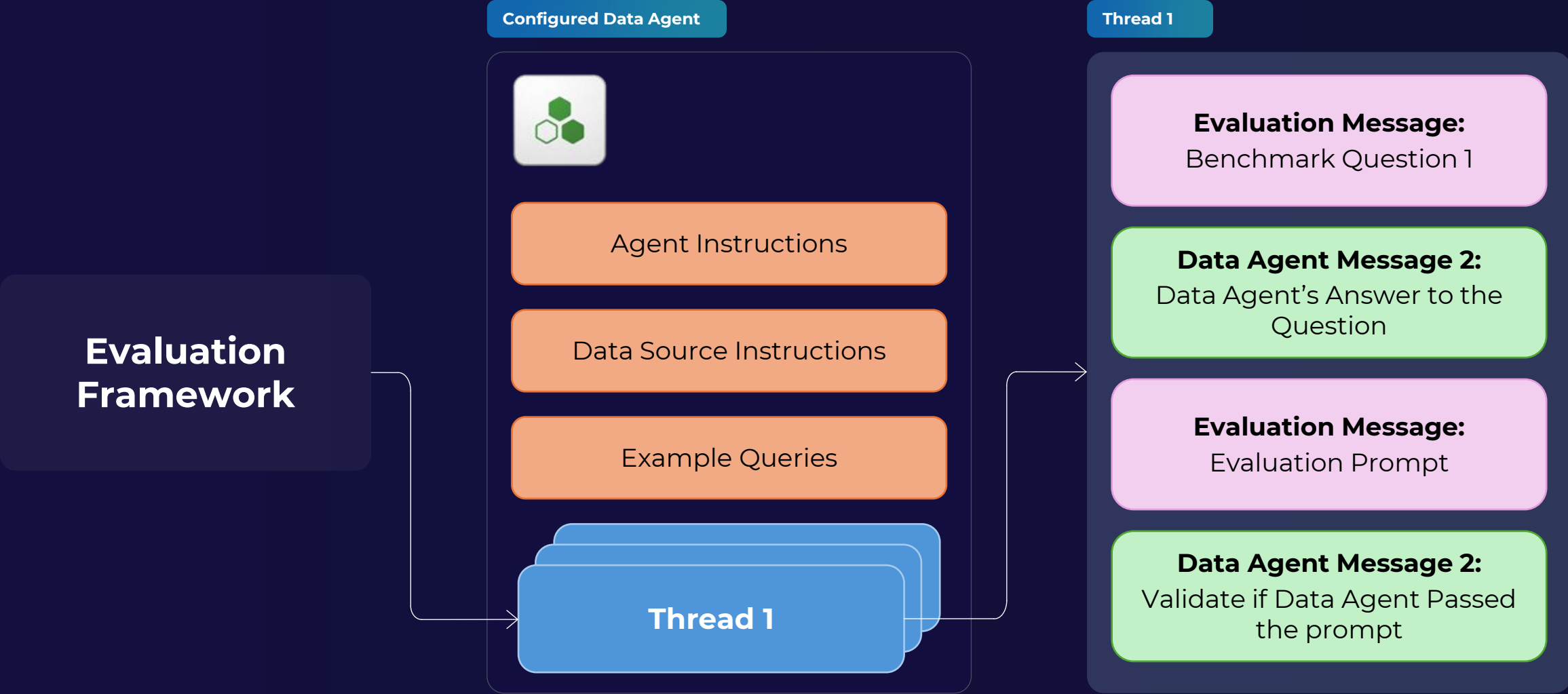
Diagnosing the Issue

	ABC Product ID	ABC Product C... ***	ABC Product Catego...	123 Product Name L...	123 Product Descrip...
1	e3e020af31d4d89d2...	beleza_saude	health_beauty	58	1987
2	c5d8079278e912d7...	beleza_saude	health_beauty	60	330
3	36555a2f528d7b2a2...	beleza_saude	health_beauty	51	1069
4	e586ebb6022265ae...	beleza_saude	health_beauty	40	1469
5	75b4372e69a42f8ae...	beleza_saude	health_beauty	22	1313
6	3569d4374a919941...	beleza_saude	health_beauty	37	1168

The Column Data Agent
Used
(Product Category)

The Column I wanted data
agent to use
(Product Category English)

Evaluation Framework



Evaluation Prompt – LLM as a Judge

Given the following query and ground truth, please determine if the most recent answer is equivalent or satisfies the ground truth. If they are numerically and semantically equivalent or satisfy (even with reasonable rounding), respond with “Yes”. If they clearly differ, respond with “No”. If it is ambiguous or unclear, respond with “Unclear”. Return only one word: Yes, No, or Unclear.

Query: {query}

Ground Truth: {ground_truth}

Evaluation is One-Shot

We evaluate on a single question, and expect the response to be correct.

Talking to Data Agent allows you to talk to the Agent. This means:

- Provide more context
- Clarify Assumptions
- Build on top of previous results

Evaluation Message:

Benchmark Question 1

Data Agent Message 2:

Data Agent's Answer to the Question

Evaluation Message:

Evaluation Prompt

Data Agent Message 2:

Validate if Data Agent Passed the prompt

Evaluation is One-Shot

There are multiple 'furniture' related products.

A User may provide additional input to the Data Agent.

Test the agent's responses

Give me sales numbers for furniture related products

I found several product categories that contain "furniture" in the name:

Category Key	English Name
moveis_decoracao	furniture_decor
moveis_escritorio	office_furniture
moveis_sala	furniture_living_room
moveis_cozinha_area_de_servico_jantar_e_jardim	kitchen_dining_laundry_garden_furniture

Please specify which of these furniture-related categories you would like sales numbers for, or let me know if you want data for all of them combined!

1 step completed

Response time: 9 sec

Evaluation is One-Shot

Question in the Evaluation:

“Give me the revenue from furniture related purchases”

 The result does not match the provided ground truth(s):

- [thread_GP6RXKhNZbEBbuofN90ABHfX](#)

100%  7/7 [01:43<00:00, 14.75s/it]

Evaluation summary per question:

index	T	F	?	%	failed_threads	question
0	1	0	0	100.0	No failed records	Compute the median product price for category beauty and health.
1	1	0	0	100.0	No failed records	Find the top city by the number of delivered orders.
2	1	0	0	100.0	No failed records	Give me the revenue from furniture related purchases

Update Configurations

Wrong data source selected

- ✓ Add **Agent Instructions** to help guide which data source to use for certain topics

Wrong filter was applied

- ✓ Add Example Queries to reinforce query logic for a given question
- ✓ Add **Data Source Instructions** to explain query behavior

Join is incorrect

- ✓ Add Example Queries to reinforce query logic for a given question
- ✓ Add **Data Source Instructions** to explain join behavior and logic

Example query not followed

- ✓ Use Evaluation SDK to validate example query quality
- ✓ Add **Data Source Instructions** to explain query behavior and logic

Evaluation Few-Shot Quality

Leverage the Data Agent SDK to Evaluate Few-Shot Quality. Across 3 Metrics:

- 1. Clarity
- 2. Mapping
- 3. Relationships

Note: This does not apply to Semantic Models for the labs

Table

+ New chart

Data Wrangler7 columns, 3 rows

Table view

DownloadFilter by keyword

	ABC Few-shot question	ABC Query (answer)	ABC Quality score	ABC Feedback (Reasoning)	ABC Clarity	ABC Mapping	ABC Relatedness	
1	Calculate the median produc...	SELECT TOP 1 -- Note that ...	yes	The natural language que...	yes	yes	yes	Inspect
2	How many orders were deliv...	SELECT c.[Customer City], -- ...	yes	The question is clear: it as...	yes	yes	yes	
3	List all orders from the fda.Sa...	SELECT * FROM [fda].[Sales] ...	yes	The question is clear: it as...	yes	yes	yes	

Evaluation Few-Shot Quality

Clarity

- Measures whether the natural language question is **clear and unambiguous**.
- Questions should be specific, include necessary metrics, timeframes, and filters, and avoid vague or multi-intent phrasing.

Relatedness

- Evaluates how closely the SQL query matches the **intent of the natural language question**.
- The SQL should return the correct metric, apply the proper filters, and match the requested granularity.

Mapping

- Checks whether **all literals in the natural language question appear in the SQL query**.
- Every number, date, or category mentioned in the question should be explicitly represented in the SQL.

Q&A

Please submit
questions at:
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dataagent-qa](https://aka.ms/dataagent-qa)



Lab 3: Evaluation

Blueprint for Microsoft Fabric Data Agent Success

Planning a Successful Data Agent Implementation

1. Scope & design

2. Get the data AI ready

3. Validate & test

4. Deploy data agent

Continuous improvement

Data Agent Implementation Plan

Planning a Successful Data Agent Implementation

**1. Scope &
design**

**2. Get the data
AI ready**

**3. Validate &
test**

**4. Deploy data
agent**

Data Agent Implementation Plan

Why Scoping Matters

Why

- Agents with laser-focused mission deliver value
- Specialized agents tie to a business outcome
- General-purpose agents with vague objectives fall short

Best practice

- Keep the agent's scope as narrow as possible while still delivering value
- Minimize data source scope

Avoid

- Overloading with too much data by maximizing data sources
- Over-engineering the agent instructions

Scoping checklist

				
Who are the primary users of the data agent?	What are the data agent's main functions?	What are the top questions users are expected to ask the data agent?	What topics are out of scope?	Are the goals of the data agent achievable with the data currently available in Fabric?
Who will use this agent, and for what exact tasks or decisions? For example, is it for analysts querying sales data or support staff searching knowledge? Define the user persona.	List the specific things the agent should do (e.g. "answer inventory levels by store", "summarize customer feedback from logs").	This will help clarify the types of queries the agent needs to support. Use these questions to evaluate whether any are too broad or not suitable for the data agent and provide feedback to refine them, so they are clearly tied to business outcomes.	Clearly note what the agent will not handle. For instance, a sales data agent won't perform HR queries. This manages expectations and prevents scope creep.	Verify that the questions in scope are answerable with available data in Fabric. Avoid "why" or open-ended questions the agent can't handle, focus on what the data can directly answer.

Examples

General purpose data agent



“Everything agent”

What are our **top-selling** products this week?

What are the top **complaint** categories in LATAM?

Which **campaigns** had the highest ROI last quarter?

What’s our **gross margin** trend YoY?

Which warehouses are running low on **stock**?



PERFORMANCE ISSUES
HARD TO MAINTAIN
NO CLEAR OWNERSHIP

CONFUSED LOGIC
LOW TRUST

Lines of Business Domains

Sales & Revenue Analytics

Inventory & Supply Chain

Marketing & Campaign Performance

Product Management & Merchandising

Customer Support & Experience

Finance & Forecasting

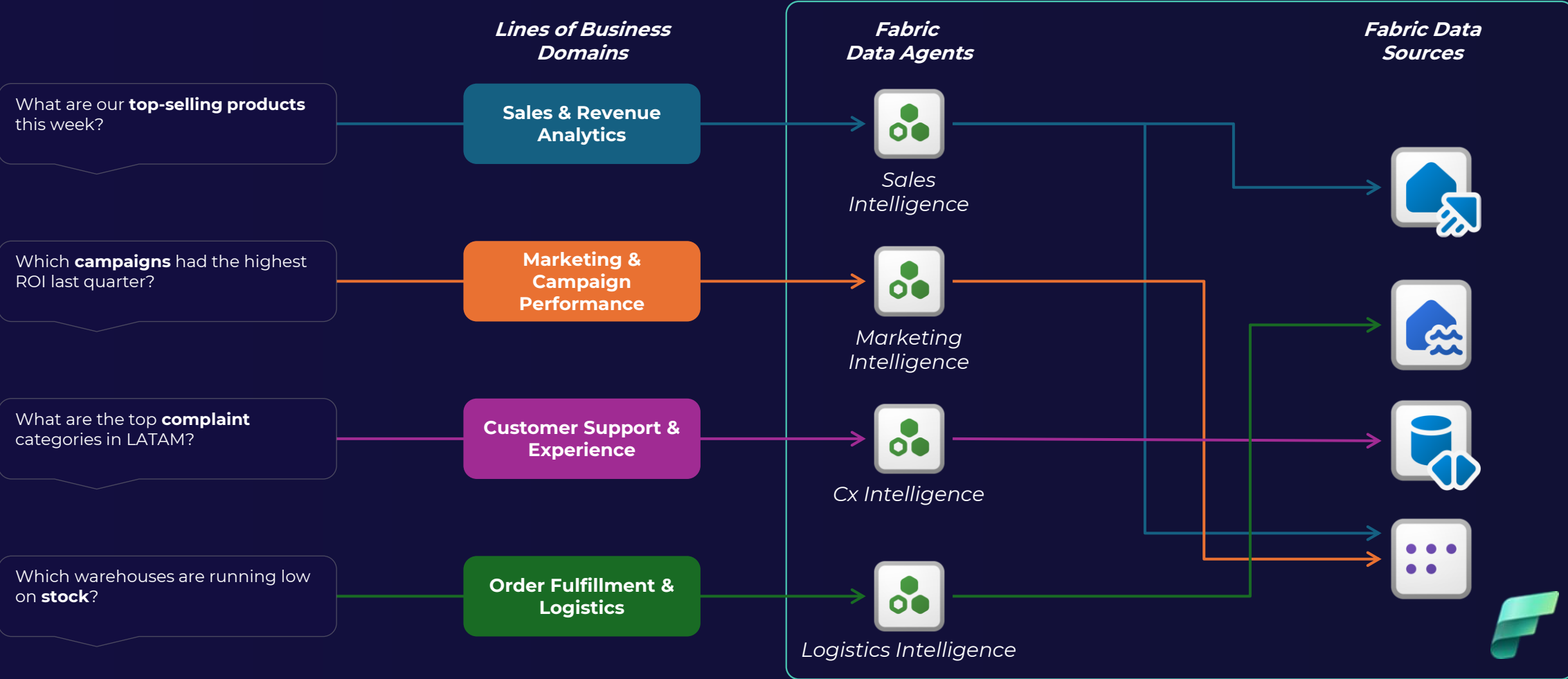
Order Fulfillment & Logistics



Fabric Data Sources

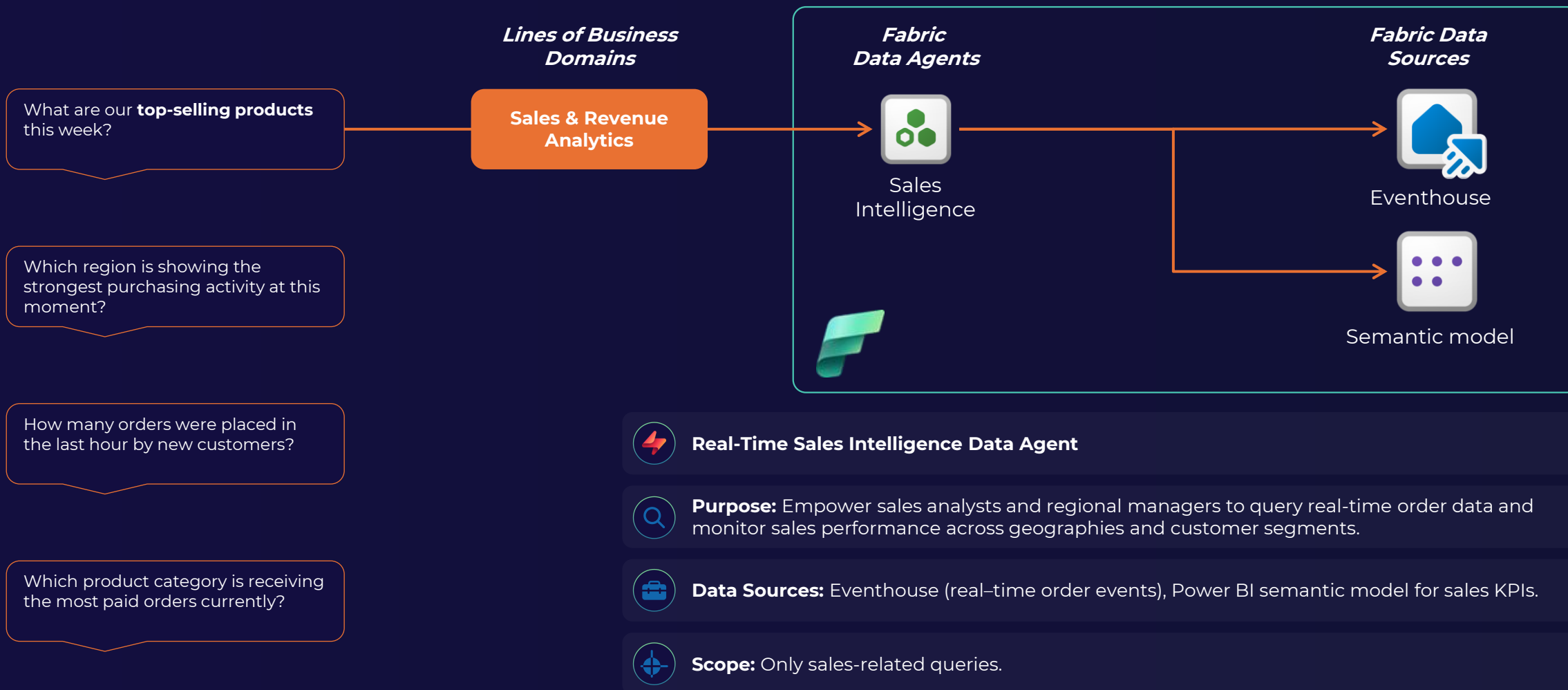


Domain-specific data agents



Real-time Sales Intelligence

Example domain-specific data agent



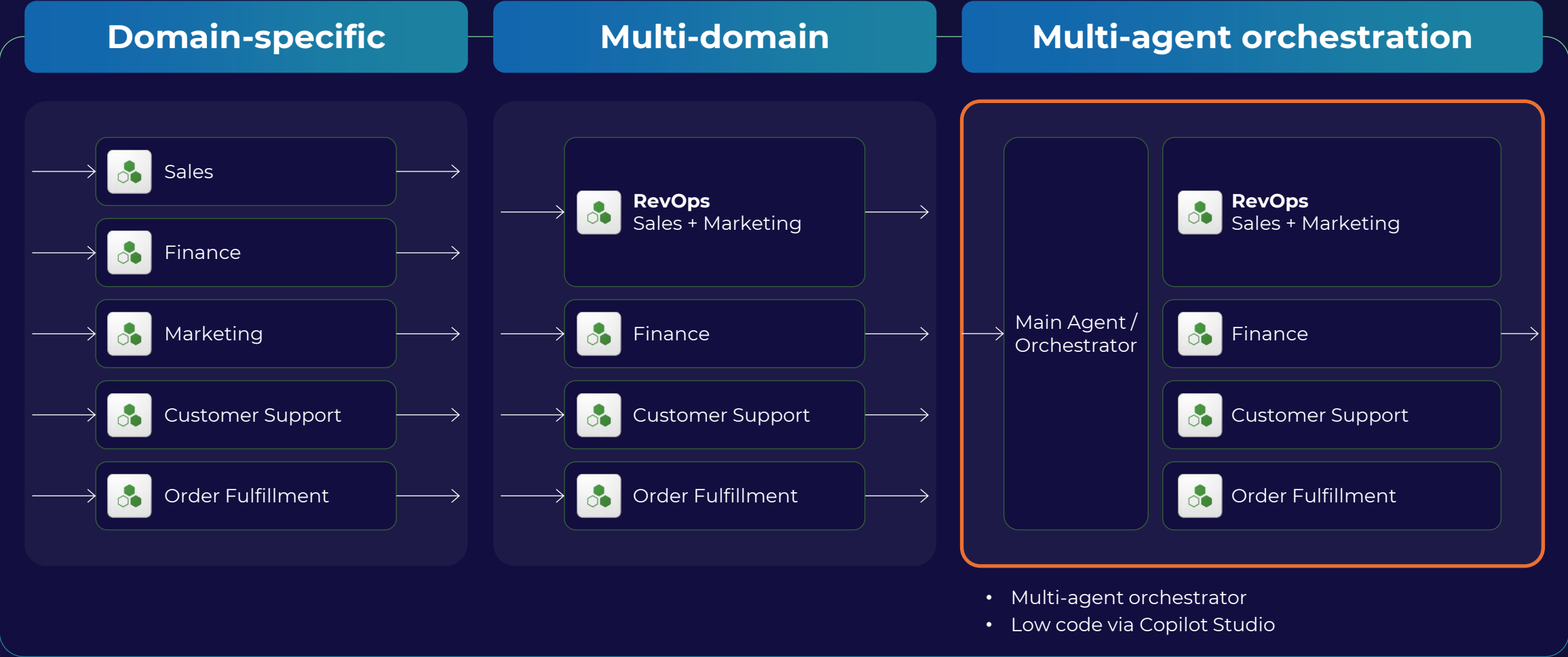
RevOps Intelligence

Example domain-specific data agent

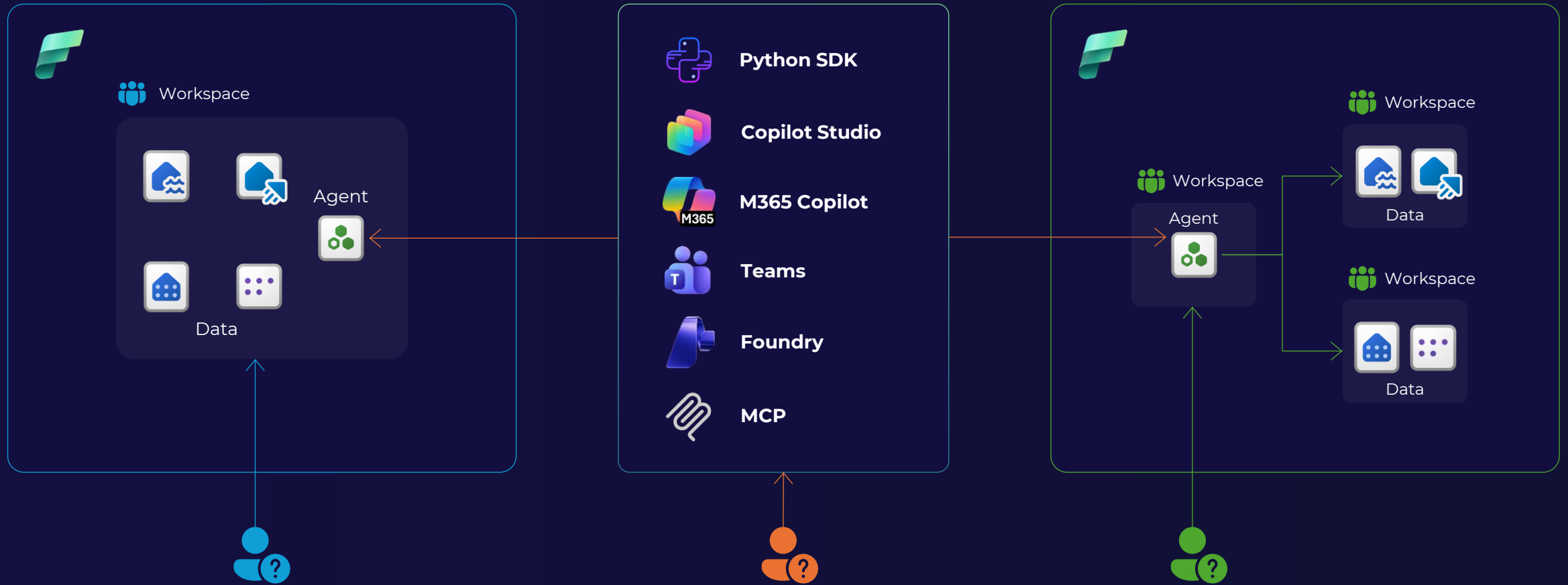


Patterns

Design patterns



Access patterns



Planning a Successful Data Agent Implementation

1. Scope &
design

2. Get the data
AI ready

3. Validate &
test

4. Deploy data
agent

Customer Implementation Plan

Assess Data Availability



Data readiness checklist

				
List data sources and assess gaps	Check data freshness	Review data quality	Evaluate volume and performance	Move data if needed
Inventory the structured data relevant to the agent's scope. This includes Lakehouses, Warehouses, Eventhouses and Power BI datasets (semantic models). For each key question in scope, ensure there is a data source/field that can answer it. If you have a gap, decide how to address it. Can it be derived? Or out of scope?	How up-to-date must the data be for this agent? Identify data that needs frequent refresh (e.g. daily sales) and ensure those pipelines are in place.	Profile the data for nulls, duplicates, or obvious errors. Aim for high-quality data in key fields. Think of the agent as unforgiving, it will surface whatever data is there, right or wrong. Also, poor metadata makes it hard for the agent to find the right info. Note which data sources have good business-friendly schema, and which may need cleanup or aliasing.	If the data source is extremely large (billions of rows), confirm that it's partitioned or indexed for performance.	Ensure needed data is in (or connected to) Fabric OneLake. Fabric Data Agents can only directly query Fabric-integrated data sources. If some data sits elsewhere plan a migration or real-time integration before including it.

Frequent questions on data source best practices



Which data source
is the best for my
data agent?



What does it
mean to have AI-
ready data?



How detailed
should our AI
instructions
really be?

Planning a Successful Data Agent Implementation

1. Scope &
design

2. Get the data
AI ready

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agent

Customer Implementation Plan

Test & optimize



Test:

- Query the agent with typical, edge, and out-of-scope queries.
- Debugging: Review agent behavior in depth (steps, generated SQL/DAX/KQL).
- Evaluate agent on factual accuracy, relevance, helpfulness, compliance.



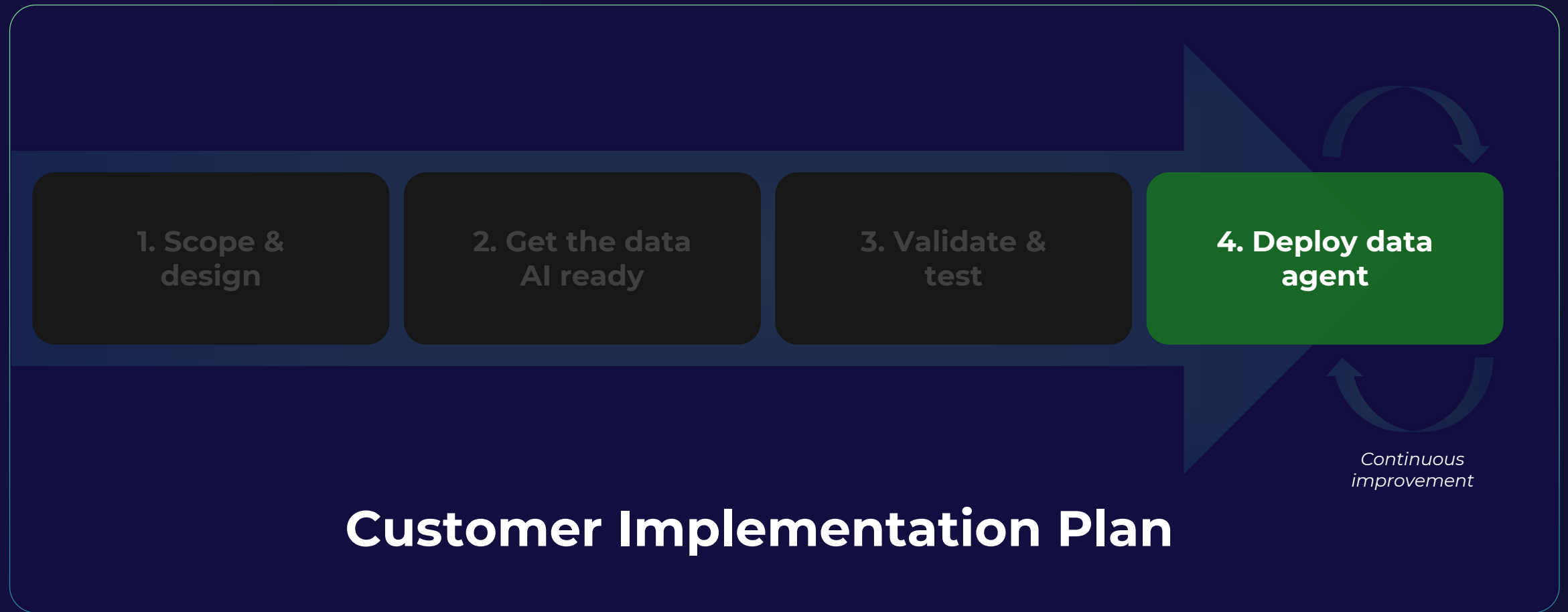
Optimization actions:

- Refined instructions or example queries.
- Adjusted data source prioritization.
- Pre-aggregate or index data for performance.



Implement CI/CD strategy

Planning a Successful Data Agent Implementation



Agent readiness checklist

			
Implement source control	Fine-tune Instructions	Secure the agent's data	Create adoption & lifecycle strategy
• All agent artifacts (instructions, data source configs, example queries) are version-controlled. • Draft vs. Published version strategy is defined. • Change log is maintained. • CI/CD process includes schema validation, regression testing, and promotion steps.	• Agent instructions include objective, data source priority, terminology, formatting, and handling rules. Instructions are tested for clarity and effectiveness. • Only relevant tables exposed. Table/column names are business-friendly. • Data source instructions include table descriptions, joins, and query logic. • Example queries added for each supported source. • Power BI semantic models include validated measures, hierarchies, and descriptions.	• RBAC and RLS configured and tested. • Sensitive data masked or excluded. • Agent queries run under user identity with correct permissions.	• Agent accessible in intended channels (Fabric, Power BI, Teams). • Quick-start guide or training provided. • Feedback loop established. • Agent owner assigned. • Maintenance plan in place (updates, feedback, schema monitoring). • Retirement criteria defined.

Planning a Successful Data Agent Implementation

1. Scope & design

2. Get the data AI ready

3. Validate & test

4. Deploy data agent

Continuous improvement

Data Agent Implementation Plan

Q&A

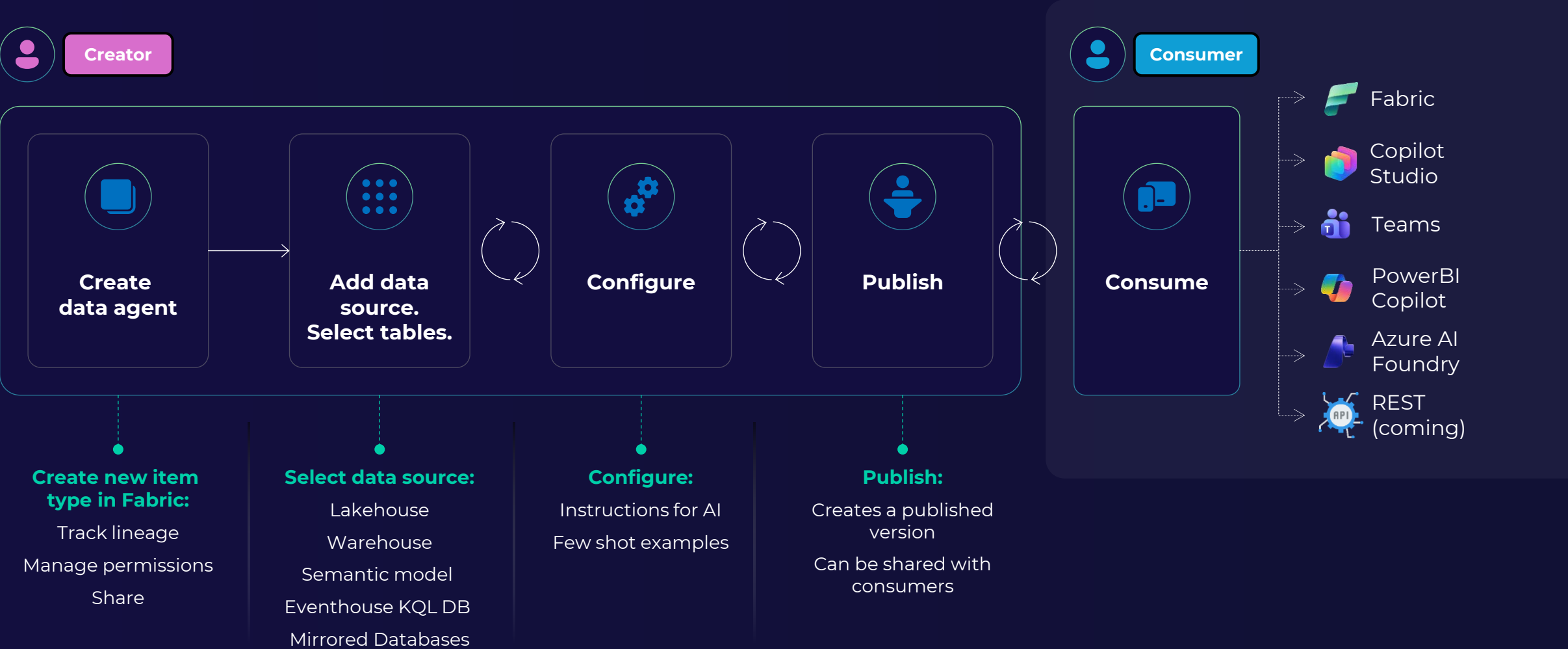
Please submit
questions at:
[https://aka.ms/
dataagent-qa](https://aka.ms/dataagent-qa)



Consumer Experiences in Data Agent

Data Agent end-to-end flow

End-to-end scenario

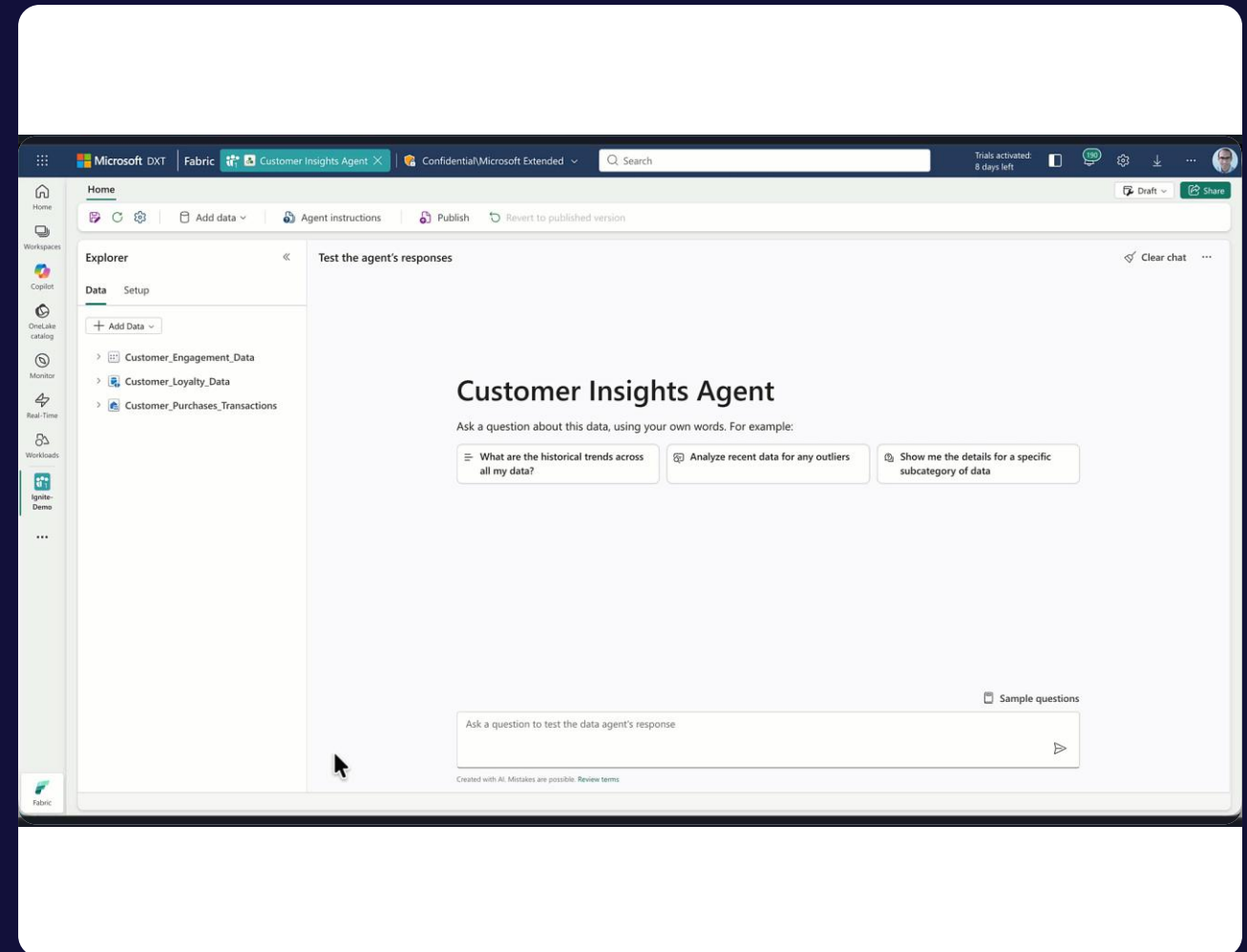


Fabric data agent integration with M365

Fabric data agents can reason over and synthesize data in OneLake for insights

AI creators can enrich M365 Copilot with Fabric data agents' data domain expertise

Security permissions, including RLS and CLS, are always respected



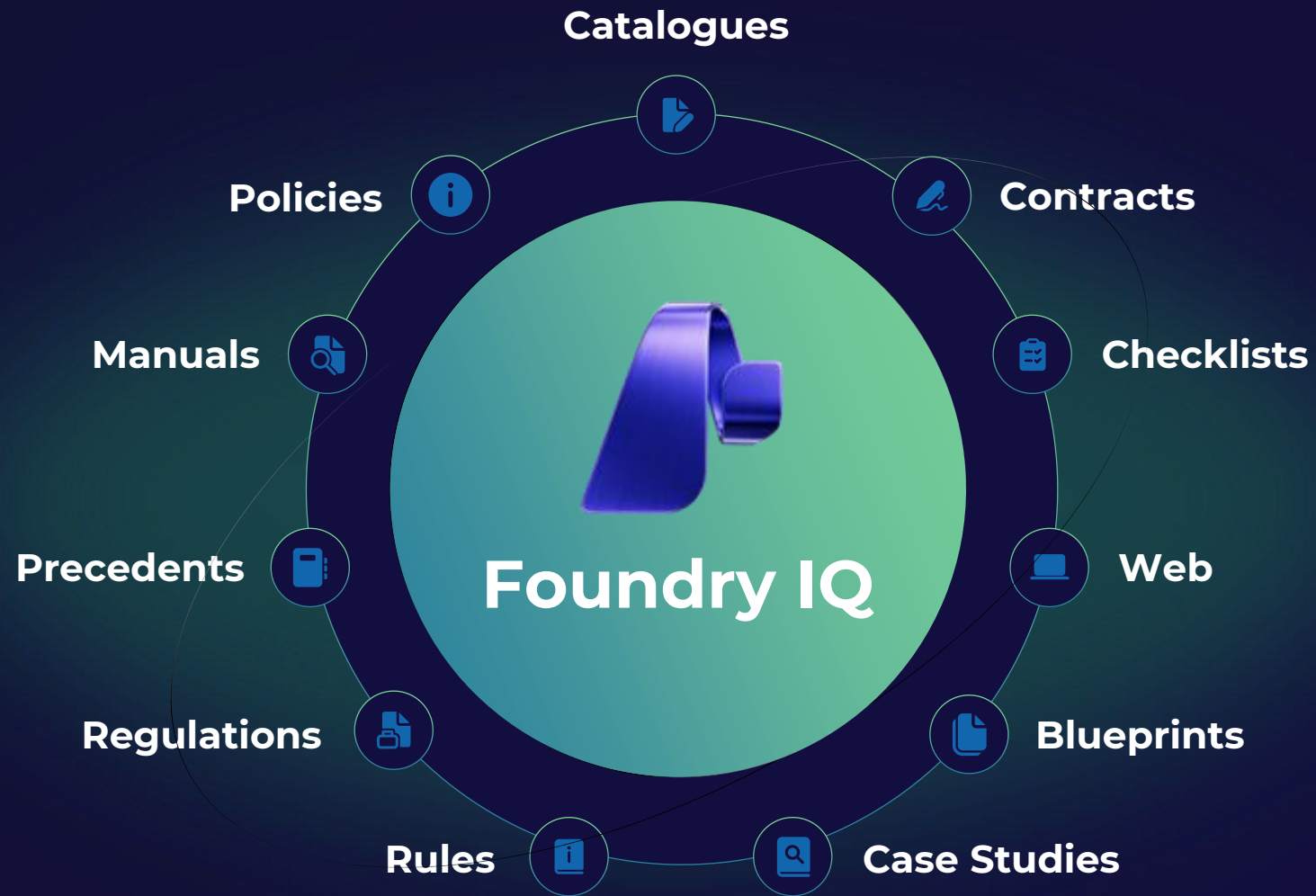
M365 – Data Agent Demo

Q&A

Please submit questions at: aka.ms/dataagent-tutorial-qa

Please provide feedback on the tutorial:
aka.ms/dataagent-tutorial-feedback

Unstructured Data in Data Agent



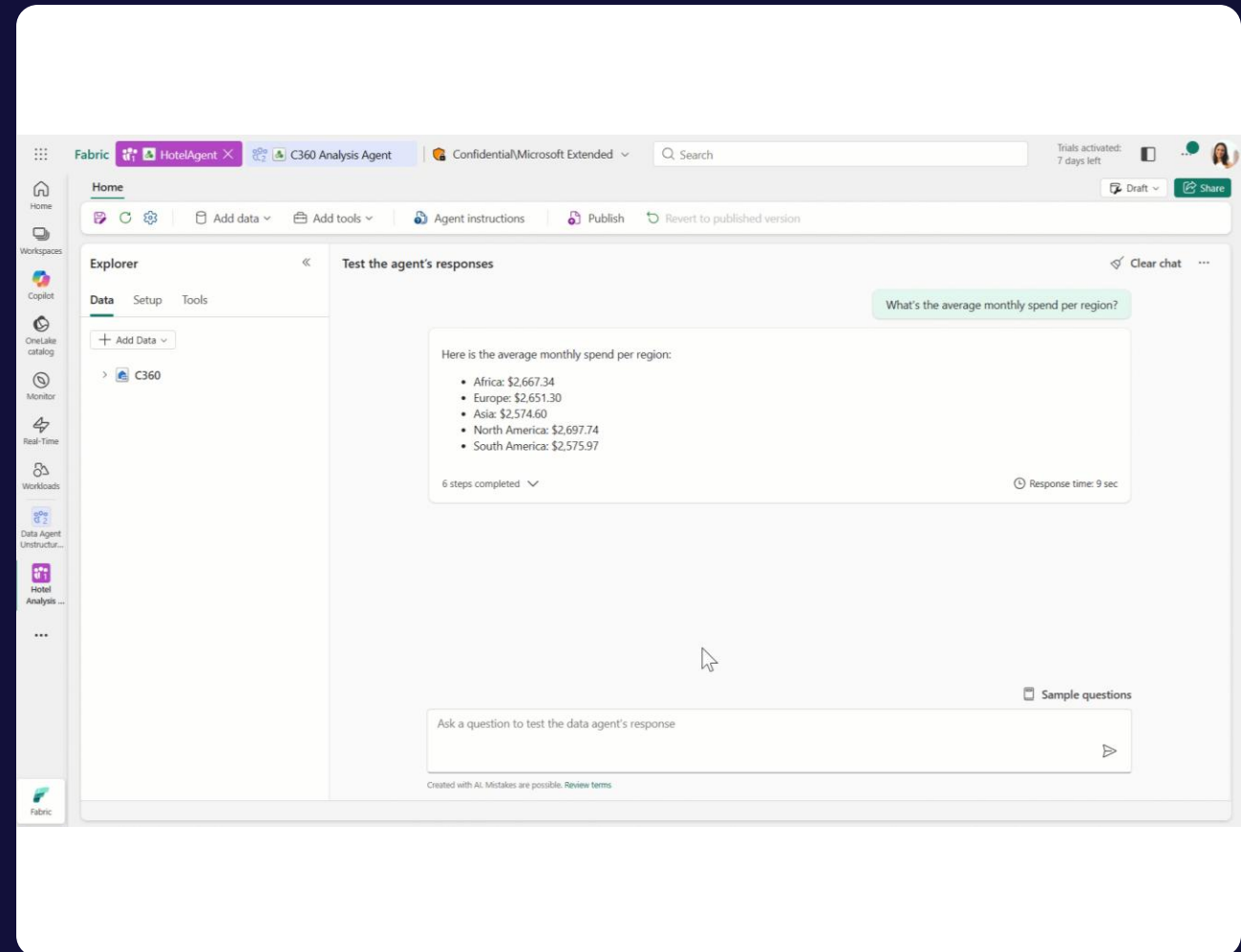
Microsoft Foundry: Context on policies, procedures, and institutional knowledge

Bring your own Foundry AI Search Index

Bring your own Foundry AI Search Index to a Fabric Data Agent

Use the built-in AI Search tool for custom indexes

Ask natural language questions over your AI Search index and join with other data sources



-
- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Search management
- Indexes
- Indexers
- Data sources
- Aliases
- Skillsets
- Debug sessions
- Settings
- Premium features
- Knowledge Center
- Keys
- Scale
- Search traffic analytics

Add index ▾

Refresh

Delete

Filter by name...

Name ▾	Document count ▾	Vector index quota ... ▾	Total storage size ▾	
custom-skillset	7,151	42.28 MB	155.48 MB	...
customer-stories	42	255.73 KB	1.57 MB	...

Lab 4: Unstructured Data

Auditing Data Agent Interactions

Viewing Prompts

“As an admin, Can I see what my users are asking Data Agent?”



Fabric does not natively offer an option to review a user's prompts or responses from Data Agent



Purview DPSM (Data Posture Security Management) for AI does!

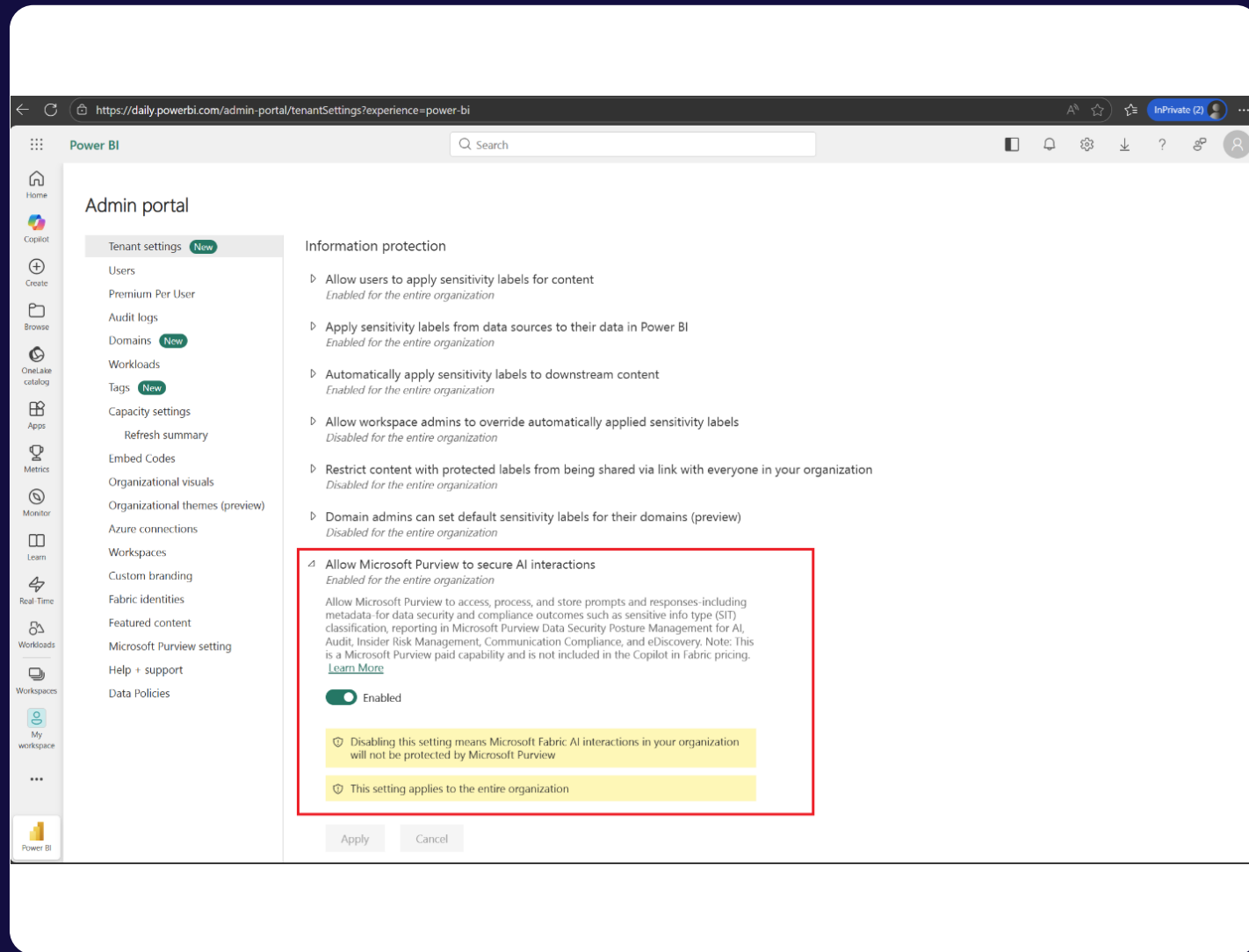
- [Security and Governance for the AI era](#)
- Requires separate licensing for Purview! ([docs](#))
- Collection Policy “DSPM for AI-Capture interactions for Copilot experiences”



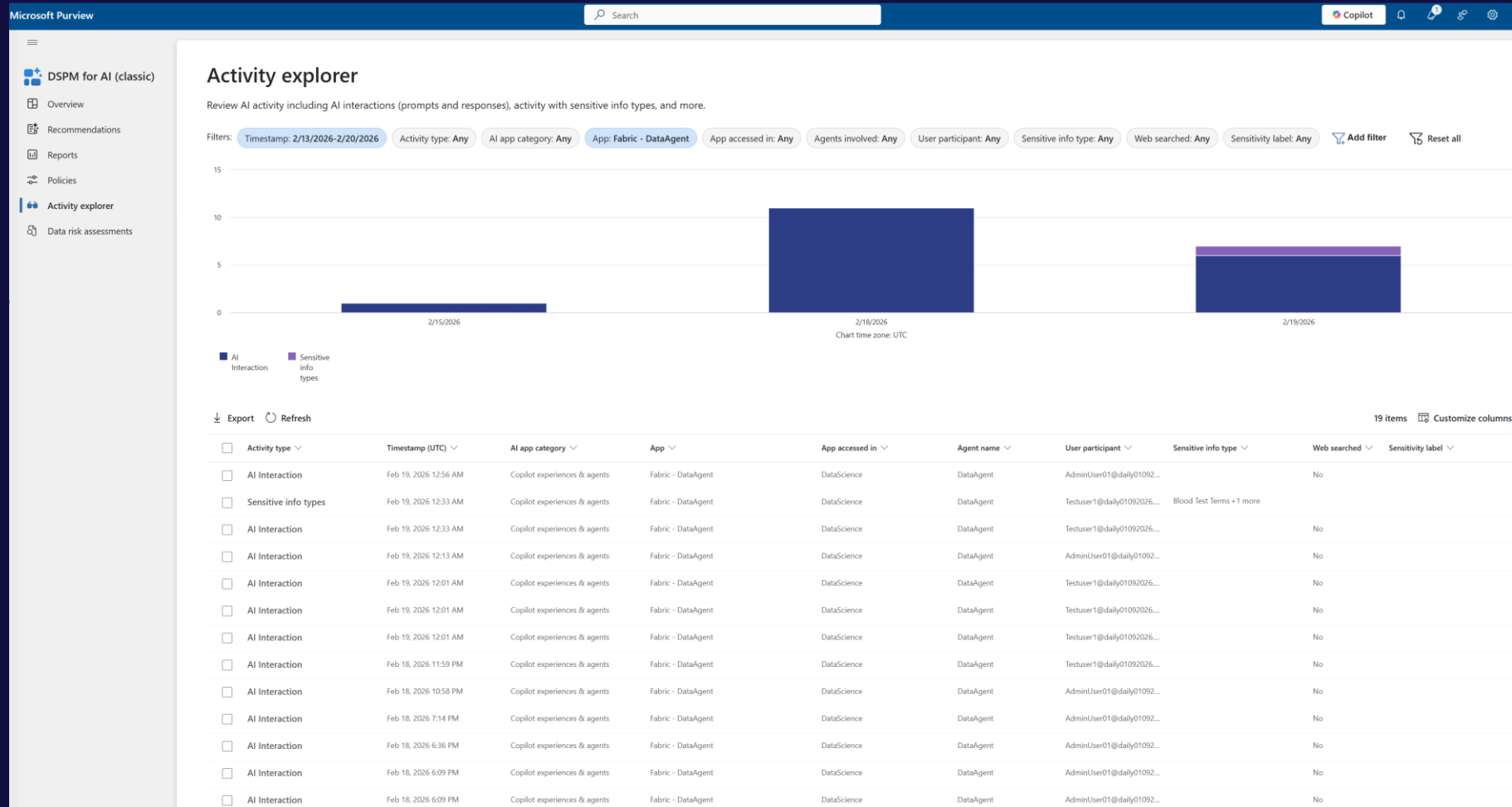
Tenant admins with Purview licenses can allow Microsoft Purview to secure AI interactions by enabling this setting within Fabric



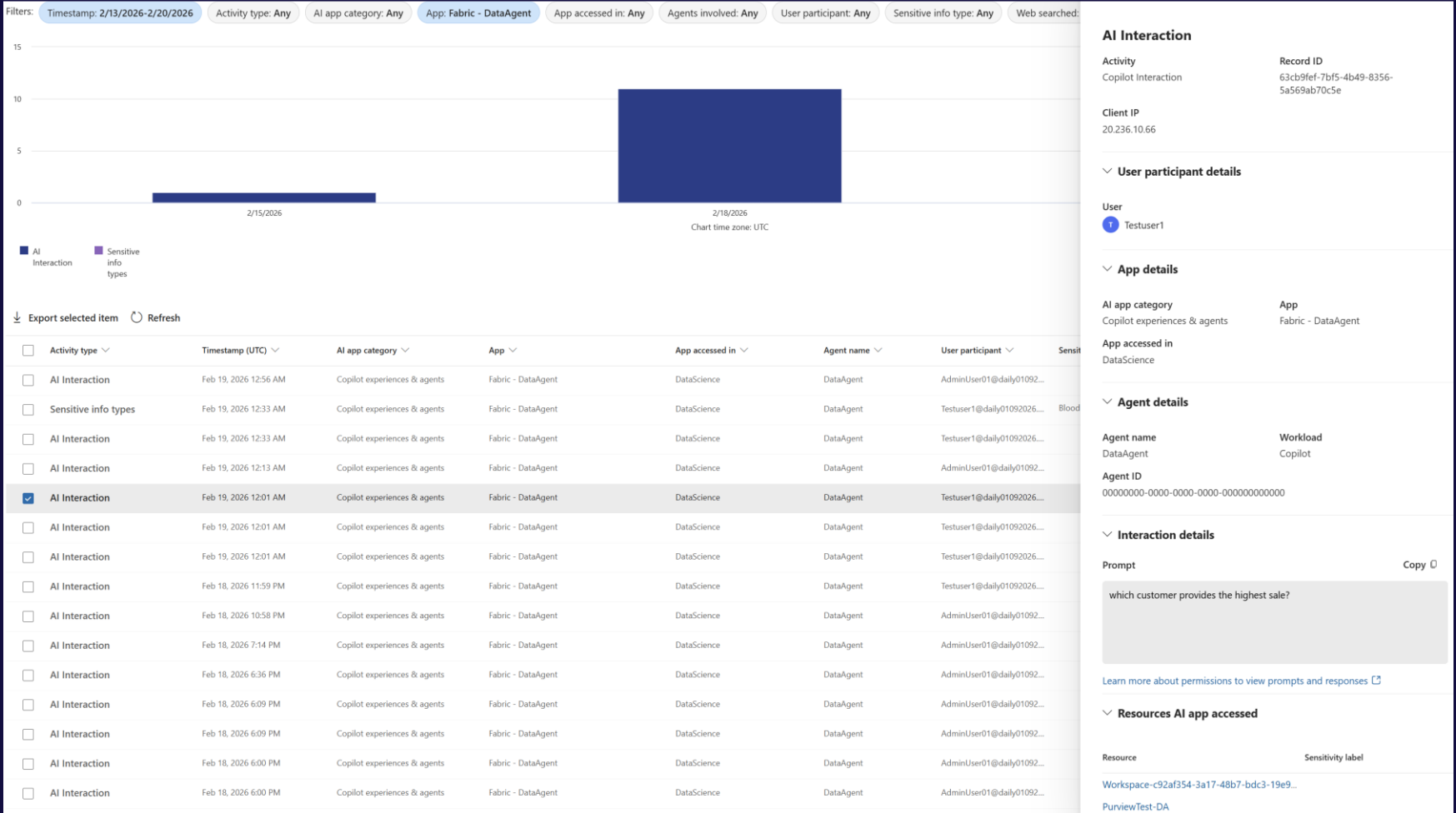
When a user performs an action within Fabric Copilots or Data Agent, an audit log entry will be created



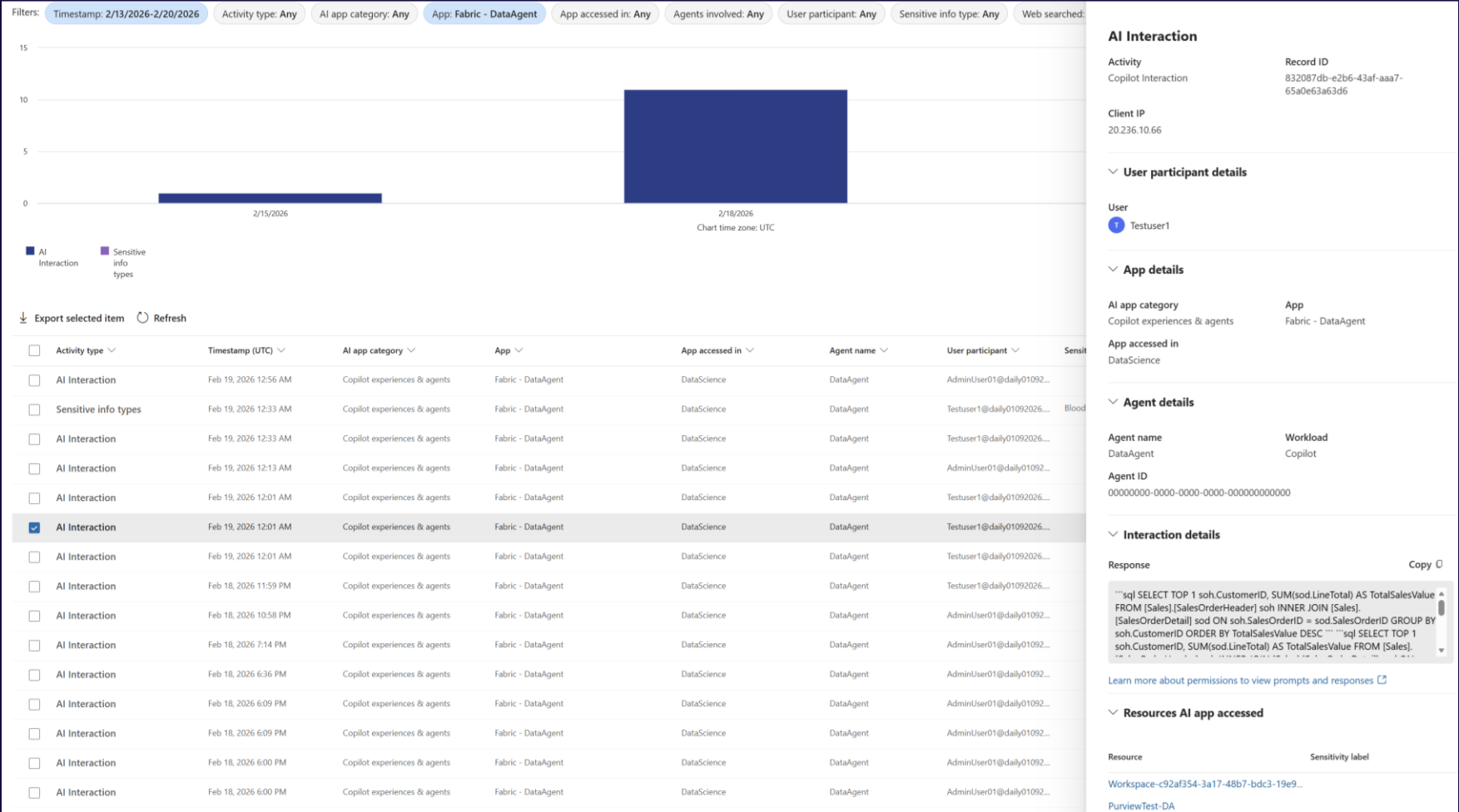
Activity Explorer – DSPM for AI



Data Agent Prompt



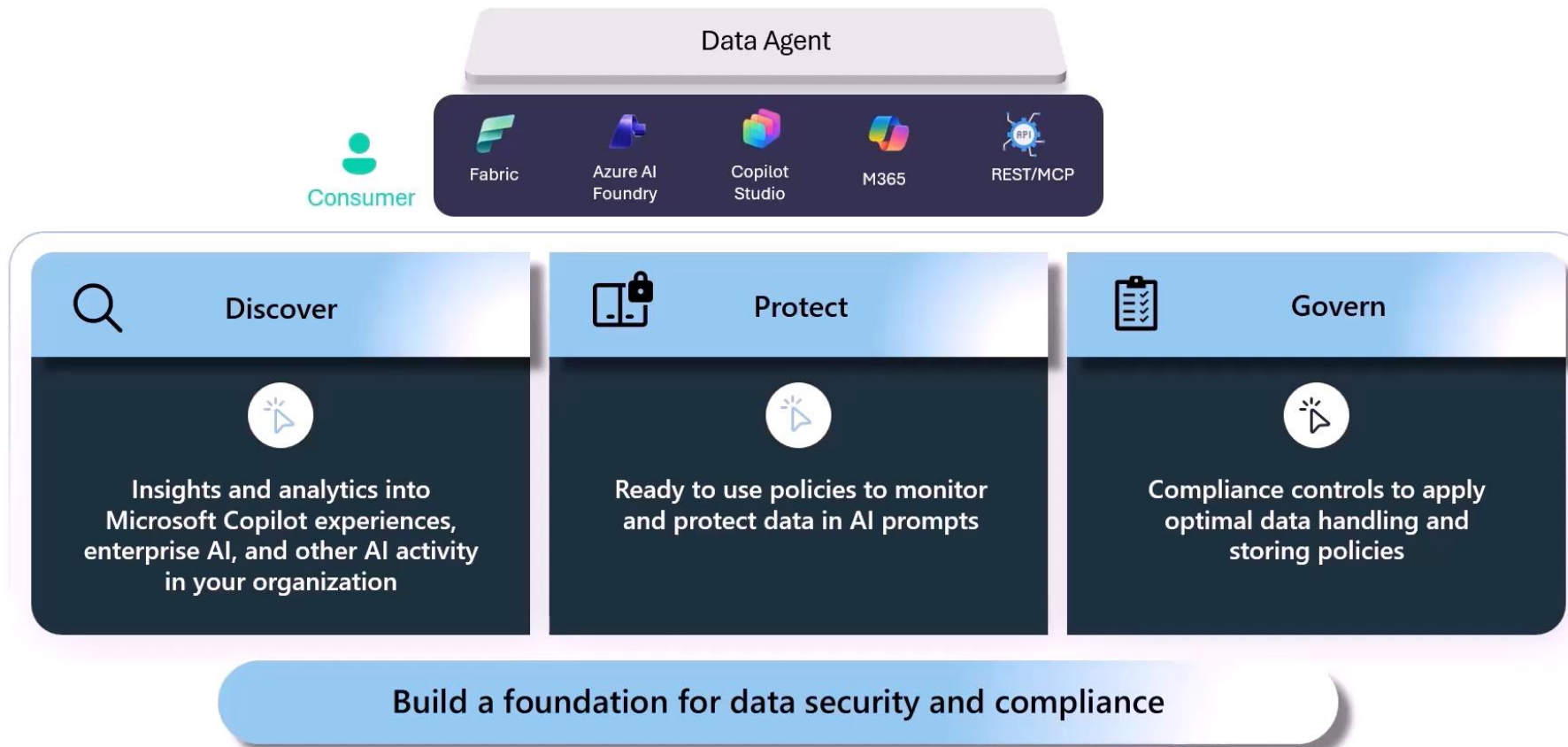
Data Agent Response



Demo

For Fabric Data Agents

Microsoft Purview discovers, protects, and governs interactions data



Don't Miss our other session throughout the conference!

Date	Time	Session Title	Speaker(s)
Wednesday, Mar 18, 2026	1:45 PM – 2:45 PM	CORENOTE: The Future of AI in Microsoft Fabric: Data Agents and Beyond	Nellie Gustafsson; Markus Weimer
Wednesday, Mar 18, 2026	3:05 PM – 4:05 PM	Unlocking Copilot in Fabric: Administration, Governance, and Beyond!	Dan English; Sandeep Pawar
Thursday, Mar 19, 2026	8:00 AM – 9:00 AM	AI Answers, Done Right: When to use Power BI Copilot vs. Fabric Data Agents	Sara Vredevoogd; Carly Newsome
Thursday, Mar 19, 2026	10:10 AM – 11:10 AM	Building and Deploying Smarter Data Agents in Microsoft Fabric	Misha Desai; Amir Jafari
Thursday, Mar 19, 2026	2:00 PM – 3:00 PM	Expert vs. AI: The Power BI Turing Test	Sujata Narayana; Sid Vemuri
Thursday, Mar 19, 2026	2:00 PM – 3:00 PM	Powering Data with AI Functions in Fabric	Virginia Roman; Rana Singh
Friday, Mar 20, 2026	11:30 AM – 12:30 PM	Reporting with action: Visualizations with translytical task flows in Power BI and Fabric	Zoe Douglas; Denyse Niwenshuti
Friday, Mar 20, 2026	2:00 PM – 3:00 PM	Beyond Monitoring: AI-Driven Spark Optimization in Microsoft Fabric	Anu Venkataraman; Virginia Roman
Friday, Mar 20, 2026	2:00 PM – 3:00 PM	Forget Prompts: Building Declarative AI Pipelines in Microsoft Fabric	Sandeep Pawar

Q&A

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questions at:
[https://aka.ms/
dataagent-qa](https://aka.ms/dataagent-qa)



Please Provide feedback
for this session:

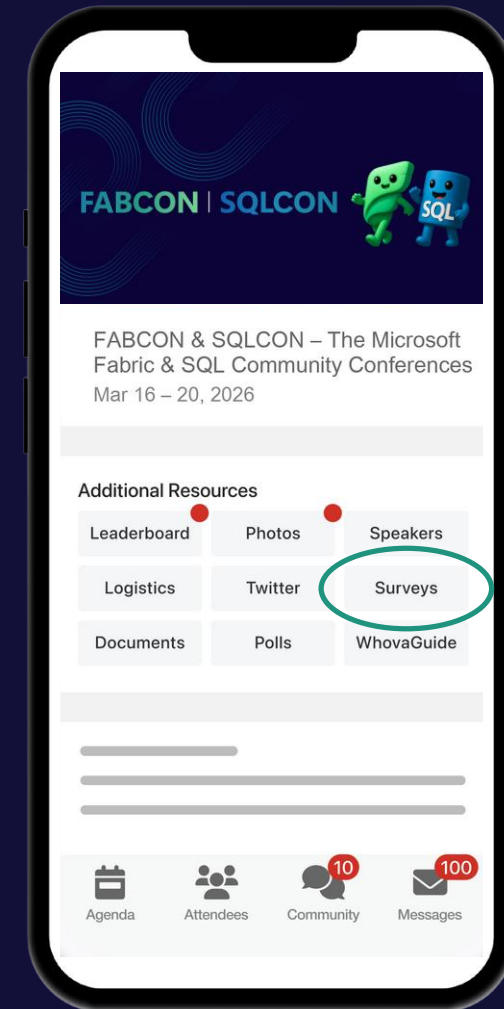
[https://aka.ms/
dataagent-tutorial-
feedback](https://aka.ms/dataagent-tutorial-feedback)



How was the session?



Complete Session Surveys in
Whova for your chance to WIN PRIZES!



End of Tutorial